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### **Integrated controller with sensors for electromechanical biasing member**

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#### **Abstract**

An electromechanical biasing member for moving a closure panel between an open and a closed position relative to a motor vehicle body is provided. The electromechanical biasing member includes a housing having an inner cavity for housing an electric motor having a motor shaft and a rotary output member rotatable relative to the housing. An extension member within the housing and engageable with the rotary output member moves the closure panel between the open and closed positions in response to driving the electric motor. Also provided is a control unit positioned within the inner cavity and configured to drive the electric motor, the control unit including at least one controller board juxtaposed to the electric motor, and a plurality of control components mounted to the at least one controller board and configured to supply a drive signal to the electric motor and receive signals representative of operation of the electric motor.

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## **Background/Summary**

CROSS-REFERENCE TO RELATED APPLICATION (1) This application is a continuation of U.S. patent application Ser. No. 17/018,772, filed Sep. 11, 2020, entitled “INTEGRATED CONTROLLER WITH SENSORS FOR ELECTROMECHANICAL BIASING MEMBER”, which is a continuation of U.S. patent application Ser. No. 15/991,289, filed May 29, 2018; which claims the benefit of U.S. Provisional Patent Application No. 62/645,282, filed Mar. 20, 2018; and also claims the benefit of U.S. Provisional Patent Application No. U.S. 62/512,468, filed May 30, 2017, each of which are incorporated herein by reference in their entirety.

### **FIELD**

(1) The present invention relates to electromechanical members for closure panels of a vehicle. In particular, the present invention relates to electromechanical members having integrated electric motors and controller components.

### **BACKGROUND**

(2) Vehicles are equipped with a variety of closure panel, such as a lift gate, which is driven between an open position (position 2) and a closed position (position 1) using an electric drive system. Vehicle closure panels can employ struts to assist the vehicle operator to open the closure panel, close the closure panel, and help maintain the closure panel in the open position or in an intermediate hold position (third position hold). Typically, the struts can be biased and can also be automatically controlled via an electric motor of the electric drive system. These struts are important in terms of both convenience and safety, because without them, vehicle operators can risk injury when entering or exiting the vehicle via the closure panel, e.g. when loading or unloading the vehicle.

(3) Therefore, it is desirable to employ motorized struts with vehicle closure panels for ease of operation. However, available space is typically limited in the vicinity of strut mounting on the vehicle frame. Current motorized strut designs utilize motors and/or motor controllers mounted on the exterior of the strut housing which may increase or alter the strut outer envelope, which is problematic due to space constraints. Further, current designs can also position motor controllers remote from the electric motors (distanced from the strut housing) in order to address the space constraints. However, remote placement of the motor controllers is also problematic due to signal noise considerations for position/obstacle sensing during operation of the strut and motor electric drive system.

### **SUMMARY**

(4) With respect to at least one of the above presented disadvantages, it is an advantage of the present invention to position electronic control circuitry and corresponding motor internal to a housing of an electromechanical basing member. Internal positioning can facilitate a reduction in packaging space requirements for the electromechanical biasing member, as compared to externally mounted control circuitry, and/or provide for desired sensing signal reception and processing quality.

(5) A first aspect provided is an electromechanical biasing member for moving a closure panel between an open position and a closed position relative to a motor vehicle body, the electromechanical biasing member comprising: a housing comprising an inner cavity, the housing for connecting to one of the closure panel and the motor vehicle body; an electric motor positioned

within the inner cavity, the electric motor comprising a motor shaft; a rotary output member within the housing and operatively coupled to the motor shaft, the rotary output member rotatable relative to the housing in response to a rotation of the motor shaft; an extension member within the housing and engageable with the rotary output member and for connecting to the other one of the closure panel and the motor vehicle body, the extension member configured to translate a rotation of the rotary output member into a linear motion of the extension member relative to the housing to move the closure panel between the open position and the closed position; and a control unit positioned within the inner cavity and configured to drive the electric motor, the control unit comprising: at least one controller board juxtaposed to the electric motor; at least one computing device mounted to the at least one controller board and configured to execute instructions to generate a control command to drive the electric motor, the at least one computing device comprising at least a data processor and a memory storing the instructions, the memory being coupled to the data processor; and a plurality of control components mounted to the at least one controller board the plurality of control components configured to supply a drive signal based on the control command to the electric motor and receive signals representative of operation of the electric motor.

(6) A further aspect provided is a method for operating an electromechanical biasing member for moving a closure panel between an open position and a closed position relative to a motor vehicle body, the electromechanical biasing member having both an electric motor comprising a motor shaft and a control unit positioned adjacent to one another within an inner cavity of a housing of the electromechanical biasing member, the control unit configured to drive the electric motor, the method comprising the steps of: receiving a command signal by the control unit, the control unit having at least one controller board juxtaposed to the electric motor; processing the command signal by at least one computing device mounted to the at least one controller board to generate an actuation signal to drive the electric motor, the at least one computing device comprising at least a data processor and a memory storing the instructions, the memory being coupled to the data processor; and sending by the control unit the actuation signal to the electric motor; wherein the at least one controller board has a plurality of control components mounted thereto, the plurality of control components configured to send the actuation signal and to receive a signal representative of operation of the electric motor.

(7) A still further aspect provided is where a plurality of control components for an electromechanical biasing member further comprise one or more sensors mounted to at least one controller board, wherein the one or more sensors comprise at least one sensor configured to detect at least one of the rotation of the motor shaft and the rotation of the rotary output member.

(8) A still further aspect provided is a control system controlling the operation of an electromechanical strut for moving a closure panel between an open position and a closed position relative to a motor vehicle body, the electromechanical strut comprising a housing comprising an inner cavity, and an electric motor positioned within the inner cavity, the control system comprising: a module associated with a vehicle latch, the module comprising a master controller configured to issue commands to move the closure panel between an open position and a closed position; and a slave controller remote from and in communication with the master controller to receive the commands, the slave controller comprising: a control unit positioned within the inner cavity and configured to drive the electric motor, the controller comprising: at least one controller board juxtaposed to the electric motor; at least one computing device mounted to the at least one controller board and configured to execute instructions to generate a control command to drive the electric motor, the at least one computing device comprising at least a data processor and a memory storing the instructions, the memory being coupled to the data processor; and a plurality of control components mounted to the at least one controller board and configured to supply a drive signal based on the control command to the electric motor and receive signals representative of an electric motor operation.

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# Description

## BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Reference is made, by way of example only, to the attached figures, wherein:
- (2) FIG. 1 is a side view of a vehicle with a closure panel assembly;
- (3) FIG. 2 is a further embodiment of the vehicle of FIG. 1;
- (4) FIG. 3 is an example biasing member as a closure panel strut of the closure panel assembly of FIG. 1;
- (5) FIG. 4 is a further embodiment of the biasing member of FIG. 3 as an electromechanical biasing member with an electronic motor assembly;
- (6) FIG. 5 is a partial cross sectional view of an embodiment of the electromechanical biasing member of FIG. 4;
- (7) FIG. 6 shows a partial side view of the electronic motor assembly of the electromechanical biasing member of FIG. 5;
- (8) FIGS. 7 and 8 are perspective views of alternative embodiments of the electronic motor assembly of FIG. 6;
- (9) FIG. 9 shows an example unassembled configuration of controller boards of the electronic motor assembly of FIG. 4;
- (10) FIG. 10 shows an assembled stacked configuration of controller boards of the electronic motor assembly of FIGS. 7 and 8;
- (11) FIGS. 11A and 11B are cross-sectional views of the electromechanical biasing member of FIG. 5, taken along the line 11'-11';
- (12) FIG. 12 is a block diagram of the controller circuit for an electronic motor assembly in accordance with an illustrative embodiment;
- (13) FIG. 13 is a circuit diagram of a processor for a control unit of the electromechanical biasing member of FIG. 5, in accordance with an illustrative embodiment;
- (14) FIG. 14 is a circuit diagram of a FET switching circuit of the electromechanical biasing member of FIG. 5, in accordance with an illustrative embodiment;
- (15) FIG. 15 is flowchart of an example operation of the electromechanical biasing member of FIG. 4 operating as a slave controller;
- (16) FIG. 16 is flowchart of another example operation of the electromechanical biasing member of FIG. 4 operating as a master controller;
- (17) FIG. 17A is a system level block diagram of an example operation of a distributed control system provided with two electromechanical biasing members one of which is operating as a master controller and the other as slave controller; FIG. 17B is a system level block diagram of an example operation of a distributed control system provided with two electromechanical biasing members operating as slave controllers to a vehicle system operating as a master controller;
- (18) FIG. 18A is a flow diagram of an example communication exchange of the distributed control system of FIG. 17A between a remote vehicle control system, a master controller and a slave controller, in accordance with an illustrative embodiment; FIG. 18B is a flow diagram of an example communication exchange of the distributed control system of FIG. 17B between a remote vehicle control system and two slave controllers, in accordance with an illustrative embodiment;
- (19) FIG. 19 is a perspective view of a latch control module viewed with a semi transparent housing, in accordance with an illustrative embodiment;
- (20) FIG. 20 is a block diagram of a distributed system architecture of an electromechanical biasing member control system employing the latch control module of FIG. 19;
- (21) FIG. 21 is a block diagram of the system modules of the latch control module of FIG. 19 operating as part of a distributed control system architecture including at least one electric motor assembly of FIG. 4;

(22) FIG. 22 is a block diagram the system modules of the electric motor assembly of FIG. 4 operating as part of a distributed control system architecture, in accordance with an illustrative embodiment; and

(23) FIG. 23 is a block diagram of a distributed control system for controlling movement of a closure panel including a latch controller and at least one electric motor assembly of FIG. 4, in accordance with an illustrative embodiment.

#### DETAILED DESCRIPTION OF EXAMPLE EMBODIMENTS

(24) In this specification and in the claims, the use of the article “a”, “an”, or “the” in reference to an item is not intended to exclude the possibility of including a plurality of the item in some embodiments. It will be apparent to one skilled in the art in at least some instances in this specification and the attached claims that it would be possible to include a plurality of the item in at least some embodiments. Likewise, use of a plural form in reference to an item is not intended to exclude the possibility of including one of the item in some embodiments. It will be apparent to one skilled in the art in at least some instances in this specification and the attached claims that it would be possible to include one of the item in at least some embodiments.

(25) In the following description, details are set forth to provide an understanding of the disclosure. In some instances, certain software, circuits, structures, techniques and methods have not been described or shown in detail in order not to obscure the disclosure. The term “control unit” is used herein to refer to any machine for processing data, including the data processing systems, computer systems, modules, electronic control units (“ECUs”), controllers, microprocessors or the like for providing control of the systems described herein, which may include hardware components and/or software components for performing the processing to provide the control of the systems described herein. A computing device is another term used herein to refer to any machine for processing data including microprocessors or the like for providing control of the systems described herein. The present disclosure may be implemented in any computer programming language (e.g. control logic) provided that the operating system of the control unit provides the facilities that may support the requirements of the present disclosure. Any limitations presented would be a result of a particular type of operating system or computer programming language and would not be a limitation of the present disclosure. The present disclosure may also be implemented in hardware or in a combination of hardware and software.

(26) Referring to FIGS. 1 and 4, an electronic motor assembly 15 is integrated within a housing 35 of an electromechanical biasing member 37 such as a spring loaded strut, for example provided as a component of a closure panel assembly 12, as further described below. The housing 35 also contains an extension member 40 used to extend from, or retract within, the housing 35 to effect the resulting location of the closure panel 14 with respect to a vehicle body 11 of the vehicle 10. For example, an extended extension member 40 results in positioning the closure panel 14 in an open state, while a retracted extension member 40 results in positioning the closure panel 14 in a closed state with respect to opening 13. It is recognized that the electromechanical biasing member 37, incorporating the electronic motor assembly 15, can be implemented as a strut (see FIG. 4 as an example type of struts). The strut can be of a biasing type (e.g. spring and/or gas charge supplying the bias). As such, via the incorporation of the electronic motor assembly 15, the strut is an electromechanical system, driven by the electronic motor assembly 15 with optionally spring and/or gas charge supplying the bias.

(27) Referring to FIGS. 1 and 2, shown is the vehicle 10 with the vehicle body 11 having one or more closure panels 14. One example configuration of the closure panel 14 is a closure panel assembly 12 including electronic motor assembly 15 (e.g. incorporated in a electromechanical biasing member 37 embodied as a strut by example) and one or more vehicle control systems 16, such as a Body Control Module (BCM), a latch control module 116, a smart latch 43, or other vehicle control system which are mounted in other locations on the vehicle body 11 (i.e. spaced apart and external/remote from the electronic motor assembly 15) and/or within the closure panel

**14**, and are coupled to the electronic motor assembly **15** via a communication path illustratively established over electrical connection(s) **110** (e.g. wired and/or wireless communication—see FIGS. **2** and **4**). For example, the electrical connection(s) **110** can be used to supply operating power to the electronic motor assembly **15**, which may be illustratively provided from a main vehicle power source **17**, such as the vehicle main battery, or other power source and/or backup energy source. For example, the electrical connection(s) **110** can be used to facilitate data and/or command signal communication between a signal source **8** (e.g. located remote to the housing **35** such as but not limited to the vehicle control system **16**) and the electronic motor assembly **15**. For example, the electrical connection(s) **110** can be used to supply both operating power to the electronic motor assembly **15** as well as facilitating data and/or command signal communication between the signal source **8** (located remote to the housing **35**) and the electronic motor assembly **15**. For example, the electrical connector(s) **110** can be configured to communicate signals associated with Local Interconnect Network protocol signals, power supply signals, and electrical ground signals. It is recognized that a battery or other type of backup power source (not shown) can be housed in the housing **35** in applications where the housing **35** footprint (e.g. dead length) is not a limiting constraint, and thus used to supply power to the electronic motor assembly **15**, such that the electrical connection(s) **110** can be used to charge the battery periodically. In one embodiment, the electrical connection(s) **110** can be used to supply both power and data/command signaling. In a further embodiment, the electrical connection(s) **110** can be used to supply power while data/command signaling is provided via wireless communication.

(28) For vehicles **10**, the closure panel **14** can be referred to as a partition or door, typically hinged, but sometimes attached by other mechanisms such as tracks, in front of the opening **13** which is used for entering and exiting the vehicle **10** interior by people and/or cargo. It is also recognized that the closure panel **14** can be used as an access panel for vehicle **10** systems such as engine compartments and also for traditional trunk compartments of automotive type vehicles **10**. The closure panel **14** can be opened to provide access to the opening **13**, or closed to secure or otherwise restrict access to the opening **13**. It is also recognized that there can be one or more intermediate hold positions of the closure panel **14** between a fully opened position and fully closed position, which can be facilitated by operation of the electronic motor assembly **15**. For example, the electronic motor assembly **15** can assist in movement of the closure panel **14** to/away from one or more intermediate hold position(s), also known as Third Position Hold(s) (TPHs) or Stop-N-Hold(s), once positioned therein. The electronic motor assembly **15** can assist with the opening and closing of the closure panel **14** in a desired manner, such as based upon a desired speed of movement, the desired third position holds, the desired anti-pinch functionality whereby the movement of the closure panel **14** is stopped from closing to avoid objects, obstacles, and limb members (e.g. fingers) from being pinched between the closure panel **14** and the vehicle body **11**, and the desired obstacle detection functionality whereby the closure panel **14** is stopped to avoid obstacles and objects from being impacted by the moving closure panel **14**.

(29) In terms of vehicles **10**, the closure panel **14** can be a lift gate as shown in FIG. **1**, or it may be some other kind of closure panel **14**, such as an upward-swinging vehicle door (i.e. what is sometimes referred to as a gull-wing door) or a conventional type of door that is hinged at a front-facing or back-facing edge of the door, and so allows the door to laterally swing (or slide) away from (or towards) the opening **13** in the vehicle body **11** of the vehicle **10**. Also contemplated are sliding door embodiments of the closure panel **14** and canopy door embodiments of the closure panel **14**, such that sliding doors can be a type of door that opens by sliding horizontally or vertically, whereby the door is either mounted on, or suspended from a track that provides for a larger opening **13** for equipment to be loaded and unloaded through the opening **13** without obstructing access. Canopy doors are a type of door that sits on top of the vehicle **10** and lifts up in some way, to provide access for vehicle passengers via the opening **13** (e.g. car canopy, aircraft canopy, etc.). Canopy doors can be connected (e.g. hinged at a defined pivot axis and/or connected

for travel along a track) to the vehicle body **11** of the vehicle **10** at the front, side or back of the door, as the application permits.

(30) Referring again to FIG. **1**, in the context of a vehicle application of a closure panel **14** by example only, the closure panel **14** is movable between a closed position (shown in dashed outline) and an open position (shown in solid outline). In the embodiment shown, the closure panel **14** pivots between the open position and the closed position about a pivot axis **18**, which is preferably configured as horizontal or otherwise parallel to a support surface **9** of the vehicle **10**. In other embodiments, the pivot axis **18** can have some other orientation such as vertical or otherwise extending at an angle outwards with respect to the support surface **9** of the vehicle **10**. In still other embodiments, the closure panel **14** can move in a manner other than pivoting, for example, the closure panel **14** can translate along a predefined track or can undergo a combination of translation and rotation between the opened and closed positions.

(31) Referring again to FIG. **1**, exemplary embodiments of the electronic motor assembly **15** (provided below) for the closure panel assembly **12** can be used as the means of open and close assistance for the closure panels **14** themselves (see FIG. **2**), or can be used in combination (e.g. in tandem or otherwise integrated) with one or more other closure panel biasing members **37** (e.g. spring loaded hinges, struts such as gas struts or spring loaded struts, etc.) that provide a primary connection of the closure panel **14** to the vehicle body **11** at a pivot connection **36** (see FIG. **1**). In general configuration of the closure panel assembly **12**, the electronic motor assembly **15** can be incorporated within a housing **35** (also referred to as lever mechanism or arm or element) used to connect the closure panel **14** as a secondary connection of the closure panel **14** to the vehicle body **11**, such that the extension member **40** and the housing **35** (of the closure panel electromechanical biasing member **37**) can be pivotally attached to the closure panel **14** at spaced apart pivot connections **36**, **38** as shown. In this manner, the end of the housing **35** pivotally connects to the closure panel **14** at pivot connection **38**. It is recognized that the housing **35** itself can be configured to contain a non-biasing element (e.g. a solid extension rod **40**) or can be configured to contain a biasing element (e.g. a gas or spring assisted extension strut containing biasing element(s)) along with the extension member **40**, as desired.

(32) Referring again to FIG. **1**, one or more optional closure panel biasing members **37** can be provided which urge the closure panel **14** towards the open position throughout at least some portion of the path between the opened position and the closed position and which assist in holding the closure panel **14** in the opened position. The closure panel biasing members **37** can be, for example, gas extension struts which are pivotally connected at their proximal end to the closure panel **14** and at their distal end to the vehicle body **11**. In the embodiment shown in FIG. **2**, there are two biasing members **37** (one on the left side (i.e. closure panel biasing member **37.sub.1**) of the vehicle **10** and one on the right side (i.e. closure panel biasing member **37.sub.2**) of the vehicle **10**), such that the closure panel assembly **12** includes the closure panel **14** and a pair of biasing members **37** acting to control the movement of the closure panel **14**. It is recognized that one or both of the biasing members **37** can incorporate the electronic motor assembly **15** within the housing **35**, i.e. thus configured as the electromechanical biasing member **37**. In one example, see FIG. **4**, the electronic motor assembly **15** is incorporated within the electromechanical biasing member **37** in order to provide a motorized version, such that the extension member **40** is actively driven by the electronic motor assembly **15** via a lead screw **140**, or drive screw **140**. The second biasing member **37.sub.2** is positioned at another side of the closure panel **14**, as a same or differently configured biasing member **37.sub.1**. In the embodiment as a differently configured biasing member **37.sub.2**, the housing **35** does not contain (see FIG. **3**) any electronic motor assembly **15** and as such this biasing member **37.sub.2** would be passively operated by motion of the closure panel **14**. In either configuration, it is recognized that during operation of the biasing member(s) **37**, the extension member **40** is either extended from, or retracted into, the housing **35**.

(33) The closure panel **14** can be opened by being powered electronically via the electronic motor

assembly **15**, for example in cooperation with one or more vehicle control systems **16**, where powered closure panels **14** can be found on minivans, high-end cars, or sport utility vehicles (SUVs) and the like. Additionally, one characteristic of the closure panel **14** is that due to the weight of materials used in manufacture of the closure panel **14**, some form of force assisted open and close mechanism (or mechanisms) are used to facilitate operation of the open and close operation by an operator (e.g. vehicle driver) of the closure panel **14**. The force assisted open and close mechanism(s) is/are provided by the electronic motor assembly **15**, any biasing member(s) **37** (e.g. spring loaded hinges, spring loaded struts, gas loaded struts, electromechanical struts, etc.) when used as part of the closure panel assembly **12**.

(34) It is recognized that the electromechanical biasing member **37** can have a lead screw **140** (e.g. rotary output member—see FIG. 4) operated actively (i.e. driven) by a motor **142** (e.g. an electrical, brushed or brushless, motor, such as a direct current (DC) electric motor) or operated passively such that the lead screw **140** is free to rotate about its longitudinal axis but is not actively driven by a motor **142** (e.g. in the event of electrical system failure of the vehicle **10**) of the electronic motor assembly **15**. It is recognized that the electronic motor assembly **15** is configured as an internal component of the electromechanical biasing member **37** (e.g. incorporated as a component positioned wholly within the housing **35**), as further provided below. The springs **66**, **68** may provide a counterbalance force to the weight of the closure panel **14** to assist with the movement of the closure panel **14**.

(35) Referring to FIG. 3, shown is a biasing member **37** (non-motorized) referred to as a biasing strut with the housing **35** (e.g. a body) having a first end **60** for connecting to the closure panel **14** (or a vehicle body/frame **11**) and a second end **62** for connecting to the vehicle body/frame **11** (or the closure panel **14**), depending upon the configuration orientation of the biasing member **37** when installed in the closure panel assembly **12** (see FIG. 1). In this configuration, the biasing member **37**, by example only, has the extension member **40** (e.g. a stator member, such as a linear drive nut **45**, slideably engageable with the rotary output member such as via a mated threads) positioned in an interior **64** of the housing **35**. A distal end **54** of the extension member **40** is coupled to the second end **62** (for example via an optional element **66**—spring) of the biasing member **37** (e.g. strut) and the proximal end **48** of the extension member **40** is coupled to the first end **60**. The extension member **40** is coupled to the lead screw **140** via the travel member **45** (for example as an integral part of or separate to the extension member **40**), such that rotation of the lead screw **140** causes travel of the travel member **45** along the lead screw **140**, to result in extension or retraction of the extension member **40** with respect to the housing **35**. As discussed in relation to FIG. 4, the travel member **45** and the lead screw **140** are coupled to one another via mated threads. As shown, the electromechanical biasing member **37** can be a strut having a resilient element of the power spring **68** for providing the counterbalance torque (T) during operation of the closure panel **14** in moving between the opened and closed positions (see FIG. 1).

(36) Referring again to FIG. 3, the travel member **45** is positioned at one end of the extension member **40**. The extension member **40** is coupled (in this example case via a mounted kicker spring **66**) to the closure panel **14** (see FIG. 1) or the vehicle body **11** at the distal end **54**. Complimentary, the extension member **40** is coupled to the vehicle body **11** or the closure panel **14** at the proximal end **48**. As such, as the extension member **40** is displaced along the longitudinal axis **41**, the attached travel member **45** is displaced along the lead screw **140**. As such, as the closure panel **14** is moved between the opened and closed positions (see FIG. 1), the position of the travel member **45** along the lead screw **140** varies, thereby providing for reciprocation of the travel member **45** along the longitudinal axis **41** of the lead screw **140**.

(37) Referring now to FIG. 4, an embodiment of the electromechanical biasing member **37** containing the electronic motor assembly **15** for the vehicle **10** is shown. The electromechanical biasing member **37** includes the housing **35** having a lower housing **112** for containing the electronic motor assembly **15** and an upper housing **114** for containing the extension member **40**

(e.g. extension shaft/rod). It is recognized that preferably the lower housing **112** and upper housing **114** are of similar lateral (e.g. diameter) dimension provided along its longitudinal length, however the housings **112,114** are shown in FIG. **4** with differing dimensions for exemplary purposes. A second pivot mount (not shown) is attached to the distal end of extension member **40**, relative to upper housing **114**, and can be pivotally mounted to the lift gate **14** of the vehicle **10**. As described below, the electronic motor assembly **15** is contained wholly within the confines of the housing **35** of the electromechanical biasing member **37**, and is thus supplied with power/data/command signals via the electrical connections **110** configured to penetrate the housing **35** through a sealed housing **35** end wall or other port (see FIG. **5**). In particular, both electronic control circuitry (e.g. ECU **144**) for the motor **142**, and the motor **142** itself, are mounted within the interior of the housing **35** without increasing the dimensions of the housing **35**.

(38) The interior of the lower housing **112** is now described in greater detail, by example. Lower housing **112** provides a (e.g. cylindrical or tubular) sidewall **122** defining a chamber **124**, or cavity. Pivot mount **121** is attached to an end wall **126** of lower housing **112** proximal to the vehicle body **11**. Upper housing **114** provides a (e.g. cylindrical or tubular) sidewall **141** defining a chamber **133** that is open at both ends. A distal end wall **128** of lower housing **112** includes an aperture **129** so that chamber **124** and chamber **133** communicate with each other. Upper housing **114** can have a smaller diameter than lower housing **112**. However, it is contemplated that lower housing **112** and upper housing **114** can also be formed as a single cylinder or frusto-cone. Other form factors for lower housing **112** and upper housing **114** will occur to those of skill in the art. Upper housing **114** can be integrally formed with lower housing **112**, or it can be secured to lower housing **112** through conventional means (threaded couplings, weld joints, etc). An electronic motor assembly **15** is seated in chamber **124** and is an integral component of the electromechanical strut **37** (e.g. situated internally in the housings **112, 114** as shown rather than situated external to the housings **112,114** as contemplated by existing state of the art designs). For example, the housing **35** can comprise one or more sections (e.g. lower housing **112**, upper housing **114**) having a cross sectional shape of circular, of rectilinear (e.g. square), etc.

(39) The electronic motor assembly **15** can include a motor **142** (e.g. a brushless electric motor, or a brushed electric motor), a clutch and planetary gearbox **150**, and a lead screw **140** (or referred to as a drive screw **140** or rotary output member powered by rotary motion of the motor **142**) which can be used to transport or otherwise guide a travel member **45** (connected to the extension member **40**) along the longitudinal axis **41** of the electromechanical biasing member **37**. Motor **142** is mounted within chamber **124** adjacent to end wall **126**. Motor **142** can be a direct current bi-directional motor. Electrical power signal line **106** and/or direction control (i.e. command control signal line **108**) for motor **142** can be provided via the electrical connections (e.g. cables or wires) **110** that connect into the electromechanical biasing member **37** through apertures or ports into the interior of the housing **35** (e.g. in end wall **126**), and are otherwise connected to operator handles, a vehicle latch, such as latch **43**, buttons and/or other vehicle control systems **16** (e.g. a potential source of the power and/or data/command signal(s)), as desired. It is recognized that the electrical connection(s) **110** can be an example of an electrical connector that penetrates the housing **35** to provide electrical communication with a computing device, such as an ECU **144** from and/or to the source **8** (e.g. vehicle control system **16**, handle, latch, button, etc.) positioned external and remote to the housing **35**.

(40) The clutch and planetary gearbox **150** are connected to an output shaft on motor **142**. The clutch and planetary gearbox **150** can provide a selective engagement between the output shaft of motor **142** and the clutch and planetary gearbox **150**. The clutch and planetary gearbox **150** can be an electromechanical tooth clutch that engages planetary gearbox when motor **142** is activated, for example. When the clutch and planetary gearbox **150** are engaged, torque is transferred from motor **142** through to the planetary gearbox **150**. When clutch is disengaged, torque is not transferred between motor **142** and planetary gearbox so that occurrence of back drive of the motor **142** can be

limited if the closure panel **14** is closed manually. For example, the planetary gearbox **150** can be a two-stage planetary gear that provides torque multiplication for lead screw **140**. Lead screw **140** extends into upper housing **114**. As such it is recognized that in the case where the electronic motor assembly **15** is present, the lead screw **140** can be driven, i.e. actively rotated by the rotary motion of the electronic motor assembly **15** coupled to the lead screw **140** via the travel member **45**. For example, the travel member **45** contains an internally facing series of threads in bore **161** that are mated to an externally facing series of threads on the extension member **40**, as desired.

Alternatively, in the case where the electronic motor assembly **15** is not present/engaged, the lead screw **140** can rotate about the longitudinal axis **41** under the influence of friction present between the travel member **45** and the lead screw **140** in the bore **161**, i.e. passively rotated by the linear motion of the travel member **45** as it rotates about the lead screw **140**.

(41) Extension member **40** resides within a (e.g. cylindrical or tubular) sidewall **154** of the housing **35** defining a chamber **156** and can be concentrically mounted between upper housing **114** and lead screw **140**. As described earlier, pivot mount **38** is attached to the distal end of extension member **40**. The proximal end of extension member **40** can be open. A nut **45** (also referred to as the travel member **45**) is mounted around the proximal end of extension member **40** relative to lower housing **112** and is coupled with lead screw **140** in order to convert the rotational movement of lead screw **140** into the linear motion of the extension member **40** along the longitudinal axis **41** of lead screw **140**. The nut **45** can include splines that extend into opposing coaxial slots provided on the inside of upper housing **114** to inhibit nut **45** from rotating as the nut **45** travels along the longitudinal axis **41**. Alternatively, the nut **45** may be configured without the splines and thus be free to rotate as the nut **45** travels along the longitudinal axis **41**, without departing from the scope of the description. An integrally-formed outer lip **164** in upper housing **114** can provide an environmental seal between the chamber **133** and the outside.

(42) A spring housing **137** can be provided in lower housing **112** and defined by a sidewall **122**, end wall **128**, and a flange **166**. Within spring housing **137**, a power spring (not shown in FIG. 4) similar to the power spring **68** as seen in FIG. 3 can be optionally coiled around lead screw **140**, providing a mechanical counterbalance to the weight of the lift gate **14**. Preferably formed from a strip of steel, power spring **68** assists in raising the lift gate **14** both in its powered and unpowered modes of the electromechanical strut **37** and alleviates the loading on the motor **142** due to the weight of the closure panel **14**. One end of power spring **68** is positioned or otherwise attached to the travel member **45** and the other is secured to a portion of the sidewall **122**. When extension member **40** is in its retracted position, power spring **68** is tightly coiled around lead screw **140** and therefore applies bias against the travel member **45**. As lead screw **140** rotates to extend extension member **40**, in concert with travel of the travel member **45** along the upper housing **114**, power spring **68** uncoils, releasing its stored energy and transmitting an axial force through extension member **40** to help raise the lift gate **14**. When power screw **140** rotates to retract extension member **40**, in concert with travel of the travel member **45** along the upper housing **114**, the power spring **68** recharges by recoiling around lead screw **140**. It is recognized that other forms of counterbalance assist configurations may be provided for assisting with the movement of the closure panel **14** by the electronic motor assembly **15** (e.g. alleviating the strain and loading due to the weight of the closure panel **14** placed upon the motor **142**).

(43) As such, in view of the above, the integrated electronic motor assembly **15** can be incorporated into a number of different electromechanical biasing member **37** form factors. One example is the strut with lead screw **140** such that the travel member **45** only travels linearly along the longitudinal axis **41**. Another example is the strut with lead screw **140** such that the travel member **45** travels both linearly along the longitudinal axis **41** and rotationally about the longitudinal axis **41** (i.e. helical relative motion).

(44) Referring to FIG. 5, shown is the electromechanical biasing member **37** with the electronic motor assembly **15** mounted in the interior of the housing **35**, such that the electronic motor

assembly **15** contains the motor **142** and a computing device, such as Electronic Control Unit (ECU) **144** for controlling operation of the motor **142**, i.e. both the motor **142** and the ECU **144** are mounted within the housing **35** containing the lead screw **140** and corresponding extension member **40** (see FIG. 4). It is recognized that both the motor **142** and the ECU **144** are mounted along the longitudinal axis **41** of the electromechanical biasing member **37**, while at the same time having the physical dimensional extents of the ECU **144** contained between interior surfaces of the housing **35**, as illustrated in FIG. 11. For example, as shown in FIG. 5, the motor **142** is within the housing **35** and adjacent to the lead screw **140** (see FIG. 4), so as to affect rotation of the lead screw **140** as directed by the ECU **144**. Further, the ECU **144** is mounted adjacent to the motor **142** and as well within the interior surface of the housing **35** (e.g. within cavity **124**). As shown in FIG. 4, the electronic motor assembly **15** is positioned in the lower housing **112** portion along with the ECU **144**, while the lead screw **140** and extension member **40** are positioned in the upper housing **114** portion. As further discussed below, the ECU **144** can be configured to function as a master controller, or as a slave controller to a master controller located externally to the housing **35** (see FIGS. 17-23).

(45) It is recognized that one embodiment shown in FIGS. 5, 12 and 17A can have the ECU **144** configured as the master controller configured to issue one or more actuations signals **144.sub.c** to actuate the motor **142** based on the command control signals **108** (or also denoted as command signals **144.sub.e** with reference to FIG. 12) received via the electrical connection(s) **110** in order to move the closure panel **14** between the open position and the closed position. As such, the electrical connection(s) **110** would be used to supply a generic indication of an open or close command **108**, as an example, issued from a vehicle control system **16**, such as the BCM, or directly from an open/close switch (e.g. a key fob **162** over wireless link **163**, an exterior closure panel handle, an interior closure panel handle, a smart latch **43**, a latch controller **116**, etc.) for receipt by the ECU **144** acting as the master controller. The command **108**, such as an open or close command, would not be directly transmitted by the ECU **144** to the motor **142**, rather the ECU **144** would be responsible for processing the open/close command **108** and then generating additional actuation signals **144.sub.c** for direct consumption by the motor **142** (see FIG. 12). In terms of master controller functionality, the ECU **144** operating as the master controller would be responsible for implementing control logic stored in a physical memory **144.sub.b** for execution by a data processor, such as processor **144.sub.a**, to generate the actuation signals **144.sub.c** (e.g. in the form of a pulse width modulated voltage for turning on and turning off motor **142** and controlling its direction and speed of output rotation of the lead screw **140**, in accordance with an illustrative example) to power the motor **142** in order to control its operation—see FIG. 12. As illustrated in FIG. 12, the ECU **144** is electrically coupled a motor driver **144.sub.g** including field-effect transistors (FETs) **148.sub.b** which are appropriately controlled (switched on/off) by the ECU **144** to generate the actuation signals **144.sub.c**. Circumstances surrounding the control of the motor **142** could include receiving sensor signals (via electronic components **148** as sensors—e.g. position sensors, direction sensors, obstacle sensors, etc.) by the master controller as the ECU **144**, processing those sensor signals, and adjusting operation of the motor **142** accordingly via new and/or modified actuation signals **144.sub.c** (e.g. adjust the period of PWM based actuation signals **144.sub.c** in the configuration where the motor **142** is responsive to supplied PWM signals). In this example, the sensor signals **148**, of sensors **148.sub.a** and the actuation signals **144.sub.c** are generated and processed internally in the housing **35** by the ECU **144**, in conjunction with the motor **142** also mounted within the housing **35**. As such, signals **108** could represent generic open/close signals, or other commands, coming from the handle(s), or other control system etc., while the actual actuation signals **144.sub.c** received by and consumed (i.e. processed) by the motor **142** would be generated by the ECU **144**.

(46) The above presented embodiment of the ECU **144** (mounted internal to the housing **35**) as the master controller is distinct from another embodiment shown in FIGS. 17B and 18B, in which the

ECU **144** acts as a slave system controller to a master controller mounted in the vehicle control system **16**, (such as latch control system **116** positioned in a housing **118** external to the housing **35**). As such, the master controller functionality supplied by the master control system **16** (e.g. the latch control system **116**, see FIG. **19** as an illustrated embodiment) is responsible for acting on a received open/close command (e.g. from a handle or key fob) and then implementing motor **142** operation via instructing the ECU **144** with lower level command details (i.e. motor speed, motor position, PWM frequency, motor mode of operation (i.e. stop motor upon manual user control detection), as but some examples) to generate the actuation signals **144.sub.c** appropriate to the circumstances of the closure panel **14** (e.g. cognizant of one or more sensor signals **144.sub.f** obtained from the electronic components **148** acting as sensors, as further discussed below). In this further embodiment, the signals **108** processed and transmitted from the ECU **144** (acting as the slave) to the master controller **16**), such as feedback signals **144.sub.h** could optionally include the sensor signals **144.sub.f** generated from the electronic components **148** acting as sensors (see FIG. **12**), which may indicate speed, position, torque output, etc. In reference to FIGS. **1** and **2**, the master controller positioned in the vehicle control system **16** (remote from the housing **35**) can be referred to as the source **8** in an illustrative embodiment.

(47) Also as shown in FIG. **5**, the electrical connection(s) **110** enter through the end wall **126** of the housing **35**, in order to be electronically coupled to the ECU **144** at one or more electrical connection locations **146** as also shown in FIG. **7**, so as to provide for integrated location of the ECU **144** completely within the interior of the housing **35** to provide for an advantage of limiting spatial (e.g. footprint) requirements (in the vehicle **10**) of the electronically controlled electromechanical biasing member **37** laterally, with respect to the longitudinal axis **41**. The need to retain/contain the spatial envelope sizing of the electronically controlled electromechanical biasing member **37**, while also providing for location of the ECU **144** in close proximity to the motor **142** and within the interior cavity **124** of the housing **35**, is considered important. The electrical connection(s) **110** are illustratively enveloped by a protective boot **99** illustratively made from a plastic or rubber like material, which also encase the end wall **126** of the housing **35** so as to provide support to the electrical connection(s) **110** and sealing from the external environment of the inner cavity **124** (defined by the opposed interior walls **153** of the lower housing **112**).

(48) As shown in FIGS. **6,7,8**, the ECU **144** can be provided as one or more controller boards, also referred to as Printed Circuit Boards **145** (PCBs), containing a number of electronic components **148** mounted thereto using techniques such as soldering for example. The PCB **145**, as generally known in the art, provides a substrate for mechanically supporting the components **148** thereto as well as providing electrical connections for the electronic components **148** with one another using conductive traces, pads and other features etched from one or more sheet layers of copper laminated onto and/or between sheet layers of a non-conductive substrate. Components **148** are generally soldered onto the PCB **145** to both electrically connect and mechanically fasten them to the PCB **145**. As shown, the ECU **144** can be implemented as the master controller, as the slave controller, or as a combination thereof. Illustratively as shown in FIG. **6**, the PCB **145** is a double sided PCB **145** having components mounted on the plane surface of first a surface **149.sub.1** facing the motor **142** and a plane surface of an opposite side **149.sub.2** facing away from the motor **142**. The PCBs **145** are illustratively circular defining a circular profile **P** and have a circular outer perimeter edge **167**, for receipt within the confines of the housing **35** inner surface **155** which is also illustratively circular and has a diameter slightly larger than the PCB **145** diameter, as shown in FIG. **11**. The PCBs **145** may be substantially circular and have a substantially circular outer perimeter, intended to mean that the radius of the PCB **145** about its center **C** is not equal at all angular points on its perimeter. Other shapes of the PCB **145** are also possible, such as square, elliptical or semi-circular, without limitation. Illustratively, notches **152** may be provided along the perimeter of the PCBs **145**, such that the PCB **145** has a substantially circular outer perimeter.

(49) For exemplary purposes, the housing **35** is shown in FIGS. **6, 7, 8** in ghosted view. Also shown

in FIGS. 7, 8, and 10 are a plurality PCBs 145 forming the ECU 144, one as a PCB 145.sub.1 connected to the motor 142 via electrically conductive legs 147, so as to position the ECU 144, for example the PCBs 145, in a spaced apart orientation along the longitudinal axis 41 (e.g. the plane surfaces of the PCBs 145 are perpendicular to the longitudinal axis 41) from the motor 142. In one embodiment, the legs 147 are equally radially distributed adjacent the perimeter of the PCB 145.sub.1, as shown in FIG. 11, but other distribution patterns are contemplated as illustrated in FIG. 10 as an example. In the illustrated embodiment of FIGS. 5 and 10, the legs 147 are illustratively provided as motor blade receptacles 147.sub.a configured to be attachable to the motor 142, for example by slidable receipt of the motor blades 147.sub.b of the motor 142 extending from the motor end wall 143, the motor blade receptacles 147.sub.a projecting from one side (i.e. surface 149.sub.1) of the PCB 145.sub.1 parallel to the longitudinal axis 41 towards the motor end wall 143. The electrically conductive legs 147 conduct the electrical signals generated by the ECU 144 (i.e. provided by the motor driver FETS 148.sub.b) to operate the motor 142, as well as conduct sensed signals 144.sub.f, such as current spikes, generated by the motor 142, representative of an electric motor operation (e.g. spikes representative of motor rotation, or stall conditions, and other operating conditions of the motor 142) to the control unit (e.g. ECU 144). The motor blade receptacles 147.sub.a are illustratively provided radially dispersed and projecting from the PCB 145 such that when connected with the motor blades 147.sub.b, the PCB 145.sub.1 is both mechanically supported and aligned with the motor 142, for example the at least one controller board 145 geometric center C is aligned co-axial with a longitudinal axis 41 of the motor shaft 157. An offset alignment is also possible depending on the configuration of the boards 145. In other words, the support members (e.g. legs 147) support the at least one controller board (e.g. PCB 145) such that the plane surface of the at least one controller board (e.g. PCB 145) is perpendicular to a longitudinal axis 41 of the motor shaft 157. As a result, the electrical connection established by the legs 147 between the PCB 145 and the motor 142 serves to provide the conductive path for the electrical actuation signals 144.sub.c from the ECU 144 positioned on the PCB 145 as will be further described herein below, while also simultaneously properly positioning and aligning the PCB 145 adjacent the motor 142, and illustratively perpendicular to the longitudinal axis 41, to thereby operate as support members. Further consequence of such connection is the proper positioning of sensors 148.sub.a, such as the hall sensors, adjacent the motor shaft 157, and in particular adjacent a magnetic pole(s) 151 mounted to the motor shaft 157 for sensing rotation of the motor shaft 157. A plurality of additional legs 147.sub.c (in ghosted view) can be used to position and support the additional parts of the ECUs 144, for example a second PCB 145.sub.2, (in ghosted view) from one another along the longitudinal axis 41, as desired, in a stacked ECU or PCB configuration as illustrated in FIGS. 8 and 10. It is recognized that the legs 147 (e.g. support members) can function as stacked ECU connectors providing electrically conductive paths (i.e. for data/command communications and/or power interconnecting one PCB 145.sub.1 to another PCB 145.sub.2 to facilitate the communications and operations there between), as desired. For example, the legs 147 can be used to mount the ECU 144 to the motor 142, so as to maintain the ECU 144 juxtaposed to the motor 142. It is recognized in FIGS. 6, 7, 8, and 11 that the PCB 145 of the ECU 144 can extend laterally with respect to the longitudinal axis 41 of the motor shaft 157, such that a lateral extent (i.e. dimension) of the PCB(s) 145 is less than a corresponding lateral dimension of the inner cavity 124 (defined by the opposed interior walls 153 of the lower housing 112) measured with respect to the longitudinal axis 41, as illustrated in FIG. 11A by the lateral extents of the PCB 145 generally represented by the dash lined P relative to the extents of adjacent opposed interior walls 153. In another embodiment, the PCBs 145 of the ECU 144 extend laterally with respect to the longitudinal axis 41 of the motor shaft 157, such that a lateral extent (i.e. dimension) of the PCB(s) 145 is less than a corresponding lateral dimension of the motor 142 measured with respect to the longitudinal axis 41, as illustrated in FIG. 11B by the lateral extents of the PCB 145 generally represented by the dash lined P relative to the lateral extents of motor 142 represented by

dashed line MP.

(50) Referring to FIG. 9 and FIG. 12, shown are example electronic components **148** on the ECU(s) **144**, for example as mounted to the boards surfaces of the PCBs **145**, located within the housing **35**, as powered/communicated with via electrical cabling **110**. The various electronic components **148** can be mounted through press-fitting or soldering for example onto one board surface **149** or onto both the opposite board surfaces **149** sensors, etc.) for detecting the rotational position/displacement of the lead screw **140**, motor shaft **157**, and/or motor **142**; motor control field-effect transistors (FETs) **148.sub.b** for directing operation of the motor **142** rotational speed/position/direction e.g. acting as a load switch to connect or disconnect a source of electrical energy (voltage and/or current) to be supplied by the ECU **144** to the motor **142**; a microprocessor or processor **148c** illustratively shown as System in Package (SIP) micro-controller logic having number of integrated circuits enclosed in a single module or package such as a microchip mounted to the PCB **145**; as well as electrolytic capacitors **148.sub.a**, resistors and other passive and/or active components for power and signaling conditioning requirements of the motor **142**, electronic sensors **148.sub.a**, and processor **148c** and other components **148** are provided. It is advantageous to have the sensors **148.sub.a** mounted on the PCB(s) **145** of the ECU **144**, illustratively on surface **149.sub.1**, located within the housing **35** and therefore adjacent to the motor **142**, as sensor performance (noise characteristics for example due to electromagnetic interference EMI) are improved when sensor signals **144.sub.f** are directly processed by the other electronic components **148** on the ECU **144**, rather than having to be communicated through the wall of the housing **35**, via electrical connection(s) **110**, to an externally located electronic controller (e.g. of the control system **16**) prior to signal processing (all or a portion thereof).

(51) Referring to FIG. 10, shown is an example of stacked PCBs **145** forming the ECU **144** adjacent to the motor **142**. Illustratively, the sensors **148.sub.a** are mounted to a first PCB **145.sub.1** on surface **149.sub.1** on the same side as the motor blade receptacles **147.sub.a**, whereas the FETs **148.sub.b** are mounted on the opposite board surface **149.sub.2** of the same PCB **145.sub.1**. The first PCB **145.sub.1** is coupled to a second PCB **145.sub.2** illustratively via the connector pins **147.sub.c**, with the second PCB **145.sub.2** having the processor **148c** mounted on a first side **149.sub.3** of the second PCB **145.sub.2** facing the second side **149.sub.2** of the first PCB **145.sub.1**, and the electrolytic capacitors **148.sub.d** mounted on the opposite board surface **149.sub.4** of second PCB **145.sub.2**. While the sensors **148.sub.a** are shown as being mounted to the first PCB **145.sub.1**, it is recognized that the sensors **148.sub.a** can be implemented as some of the electrical components **148** on the PCBs **145** or otherwise mounted off of the PCBs **145**, such as positioned in the motor **142**, or adjacent the lead screw **140**, or elsewhere, while being electrically connected to the PCBs **145**.

(52) Referring now to FIG. 12, the integrated ECU **144** of the electromechanical biasing member **37** and its interconnection with the various electronic components **148** is schematically represented. The ECU **144** can include a processor **144.sub.a** (e.g. microprocessor or processor **148.sub.c**) and a set of instructions **159** stored in the physical memory **144.sub.b** for execution by the processor **144.sub.a** to determine the actuation signals **144.sub.c** (for example, actuation signals in the form of a pulse width modulated voltage for turning on and turning off motor **142** and controlling its direction of output rotation) to power the motor **142** to control its operation in a desired manner. The memory **144.sub.b** may include a random access memory ("RAM"), read-only memory ("ROM"), flash memory, or the like for storing the set of instructions **159**, and may be provided internal the processor **144.sub.a** or externally provided as a memory chip mounted to the PCB **145**, or both. The memory **144.sub.b** may also store an operating system for general management of the ECU **144**. As such, the electrical components **148** with the PCB(s) **145** can be considered an embodiment of the control circuitry provided by the ECU **144** which operate together to form at least one computing device for processing data by a processor (e.g. processor **144.sub.a**) such as communication signals, command signals **144.sub.e**, sensor signals **144.sub.f**, feedback signals

**144.sub.h** and executing code or instructions stored in a memory (e.g. memory **144.sub.b**) and outputting motor **142** control signals and for processing other communication/control signals and algorithms and methods in a manner as illustratively described herein.

(53) Now referring to FIG. **13**, there is illustrated a circuit diagram of a processor **144.sub.a** (e.g. microprocessor **148.sub.c**) in accordance with an illustrative embodiment. The processor **144.sub.a** (e.g. microprocessor **148.sub.c**) is mounted to PCB **145** and includes a number of input/output ports **300** providing various interfaces with internal and external signals lines, such as power ports **300.sub.a** which may be illustratively connected to main vehicle power source **17** via the communications lines **110**, digital and analog hall sensor port **300.sub.b**, FET control ports **300** for providing FET control signals for controlling the switching of the FETs **148.sub.g** to generate the actuations signals **144.sub.c** to be supplied to the motor **142** via the motor blade receptacles **147.sub.a**, such as motor open signal port **301**, and motor close signal port **302**, operational amplifier ports **300.sub.d** for interfacing with current sensing circuits (not shown), programming ports **300.sub.e** for receiving programming instructions, for example from memory **144.sub.b** and testing ports **300.sub.f**, as examples. Supporting electronics such as diodes **310**, capacitors **312**, and resistors **314** are provided for conditioning signals and signal levels to be provided to and/or supplied by to the processor **144.sub.a** (e.g. microprocessor **148.sub.c**).

(54) Further, referring back to FIG. **12**, the ECU **144** can have a communication interface **144.sub.d** to receive any power and/or data/command signal(s)), such as receive control command signals **144.sub.e** from the electrical connection(s) **110** (issued by the remote/external control system **16**) and in turn to control the operation of the motor **142** in response. The ECU **144** may optionally have a dedicated power interface **144.sub.j**. Likewise, communication interface **144.sub.d** may be configured to supply power and/or data/command signal(s)), such as subcommand signals **144.sub.1** to the electrical connection(s) **110** (for transmission to external systems from the electromechanical biasing member **37**, or other electromechanical biasing members **37** when operating as a slave device). The communication interface **144.sub.d** may include one or more network connections adapted for communicating with other data processing systems (e.g., similar to ECU **144** provided on another biasing member **37** in communication) over a vehicle network or bus via, and in the illustrative embodiment over the electrical connection(s) **110** which may form part of such as bus. For example, the communication interface **144.sub.d** may be connected to a Local Interconnect Network (LIN) or CAN bus or the like network protocol, over which command signals issued by the control system **16** over the vehicle network may be received and/or transmitted. As such, the communication interface **144.sub.d** may include suitable transmitters and receivers. Thus, the ECU **144** may be linked to other data processing systems by a communication network, which electrical connection(s) **110** may form part of. The communication interface **144.sub.d** may also be of a wireless configuration capable of sensing and transmitting communication signals wirelessly, for example using RF frequencies or the like, over wireless link **163**. The input/output arrangements of the communication interface **144.sub.d** can be built into an I/O arrangement on the PCB(s) **145** of the ECU **144** for integration within the housing **35**. Optionally, it may be integrated into the microprocessor **144.sub.a**.

(55) Command signals **144.sub.e** received by the communication interface **144.sub.d** may include data related a generic or high level command to open the closure panel **14** to a certain position; to hold the closure panel **14** at this position; to fully open the closure panel **14**; to fully close the closure panel **14**; as but a list of non-limiting examples of commands. For example, a generic "CLOSE" command received by the communication interface **144.sub.d** could result in the actuation signal **144.sub.c** to drive the motor **142** at certain speeds (e.g. the ECU **144** may control the switching frequency of FETS **148.sub.g** to adjust the power allowed to be conducted to the motor **142**) over a defined path of movement from fully open, to a point/position before the fully close position where the actuation signal **144.sub.c** would be adjusted by the ECU **144** to reduce the speed of operation of the motor **142** (e.g. the ECU **144** may decrease the switching frequency of

FETS **148.sub.g** to adjust the power allowed to be conducted to the motor **142**) and stop movement of the closure panel **14** (e.g. the ECU **144** may control the FETS **148.sub.g** to stop conducting power to the motor **142**) at a predefined point/position of the closure panel **14**. For example, such a point may correspond to a position of the closure panel **14** whereat the latch **43** engages a striker **44** provided on the vehicle body **11** where it is in an aligned position of with the striker **44** to perform a cinching operation to thereby transition the closure panel **14** to the fully closed position without an operation of the motor **142**, the cinching operation involving the transitioning of the latch **43** from a secondary latched position to a primary latched position as is generally known in the art. As a result, the striker **44** provided on the closure panel **14** which is moved by the movement of the closure panel **14** into a position where the striker **44** engages the secondary position of the latch **43** to capture and maintain the striker **44** in latched engagement with the latch **43**. At such a position, the motor **142** may be deactivated so as not to interfere with the cinching operation of the latch **43**. Sensors **197** provided in the latch **43** and in communication directly or indirectly with the ECU **144**, (for example via electrical connection(s) **110**) may assist the ECU **144** to determine locally the actuation signal **144.sub.c** required to stop the motor **142** at this position. Illustratively, sensor **197**, such as an accelerometer, (see FIG. 2) may generate sensor signals to be communicated from the latch **43** to the ECU **144** via the electrical connections **110**. It is recognized that other command signals can be issued, such as to move the closure panel **14** from the fully opened to a secondary latching position whereat the vehicle latch **43** is moved into the secondary latched position in position for a cinching operation to transition the latch **43** from the secondary position to the primary latched position, and for other closure panel **14** movement operations. The processor **144.sub.a** can therefore be programmed to execute instructions as a function of the command signals **144.sub.e** transmitted and received by the communication interface **144.sub.d** as Local Interconnect Network protocol signals such as but not limited to commands for operating electronic motor assembly **15** in a mode of operation including: a position request for motion mode, a push to close command mode, a push to open command mode, a time detected obstacle mode, a zone detected obstacle mode, a full open position detected mode, a learn mode, and/or an adjustable stop position mode, as will be described herein below in more detail.

(56) Now referring to FIG. **14** in addition to FIGS. **12** and **13**, the ECU **144** is configured to interpret the command signals **144.sub.e** received at the communication interface **144.sub.d** from the external or remote system **16** and in response activate the motor driver **144.sub.g** including the FETS **148.sub.b** appropriately, for example based on a stored movement sequence or profile stored in memory **144.sub.b** and referenced (e.g. looked up in memory **144.sub.b**) based upon, at least in part, the received command signals **144.sub.e**. Such predefined stored movement sequences of the closure panel **14** may be recorded in the memory **144.sub.b**. For example, the received command signals **144.sub.e** may be a digital message encoded according to a communication protocol (e.g. a serial binary message-based protocol), the ECU **144** capable of decoding the digital message to extract the command (e.g. converts the data stream received by the communication interface **144.sub.d** as serial bits (voltage) levels into data that the ECU **144** can process). In response, ECU **144** may issue FET control signals to FET control ports **300c** to control the operation of the FETs **148.sub.b** (e.g. control the FET gates) to supply current and/or voltage to the motor blades receptacles **147.sub.a** to operate the motor **142**.

(57) The ECU **144** can be further programmed by the execution of instructions **159** to operate the motor **142** based on different desired operating characteristics of the closure panel **14**. For example, the ECU **144** can be programmed to open or close the closure panel **14** automatically (i.e. in the presence of a wireless transponder (such as a wireless key FOB **162**) being in range of the communication interface **144.sub.d**) when a user outside of the vehicle **10** initiates an open or close command of the closure panel **14**. Also, the ECU **144** can be programmed to process feedback signals **144.sub.t** from the electronic sensors **148.sub.a** supplied to the ECU **144** to help identify whether the closure panel **14** is in an opened or closed position, or any positions in between.

Further, the closure panel **14** can be automatically controlled to close after a predefined time (e.g. 5 minutes) or remain open for a predefined time (e.g. 30 minutes) based on the instructions **159** stored in the physical memory **144.sub.b**. For example, the high level generic command (e.g. **144.sub.e**) may include a command labelled, for illustrative purposes only: "Open Profile A", which may be decoded by the ECU **144** to undertake operation of the biasing member **37** to move the closure panel **14** in accordance with a sequence of operations as stored in memory **144.sub.b** including three aspects such as moving the closure panel **14** to fully open position, a hold open for a period of time (i.e. 3 minutes) after the closure panel **14** has reached the fully opened position, and a fully closing operation after a second period of time (i.e. 5 minutes) after the closure panel **14** has reached the fully opened position. For example, the high level generic command (e.g. **144.sub.e**) may include a command labelled "Open Profile B", which may be decoded by the ECU **144** to undertake similar operations of "Open Profile A" except replacing the fully closing operation with an expected manual user movement of the closure panel **14** as would be detected by the sensors **148.sub.a**. Further, the processor **144.sub.a** can be programmed to execute the instructions complementing and enhancing the functionality of the closure panel **14** locally of received profile command, for example executing a sub-profile operating mode, based on received signals **144.sub.f** from the electric motor **142** representative of an electric motor **142** operation selected from operations such as but not limited to: an electric motor speed ramp up and ramp down operating profile, an obstacle detecting mode for detecting obstructions of the pivotal closure member between an open position and a closed position, a falling pivotal closure member detection mode, a current detection obstacle mode, a full open position mode, a learn completed mode, a motor motion mode, and/or an unpowered rapid motor motion mode.

(58) As another illustrative example of locally controlled operation of the biasing member **37**, a manual override function is described. One or more Hall-effect sensors **148.sub.a** may be provided and positioned within the housing **35**, as illustrated in FIG. **6** the Hall-effect sensors **148.sub.a** are positioned on the PCB **145** adjacent to the motor shaft **157**, to send a signal, such as an analog voltage time varying signal depending of the change in magnetic field detected by the Hall-effect sensors **148.sub.a**, representative of operation (e.g. motor output shaft **157** rotation(s)) of the electric motor **142** to ECU **144** that are indicative of rotational movement of motor **142** and indicative of the rotational speed of motor **142**, e.g., based on counting signals from the Hall-effect sensor **148.sub.a** detecting a target (i.e. magnet **151**) on the motor output shaft **157**. In situations where the sensed motor **142** speed is greater than a prestored expected threshold speed, stored in memory **144.sub.b** for example, and where a current sensor (in the case where ripple counting is employed to determine the operation of the motor **142**, such as to determine the position of the motor **142**) registers a significant change in a current draw, the ECU **144** may determine that a user is manually moving the closure panel **14** while motor **142** is also operating to rotate the lead screw **140**, thus moving the closure panel **14** between its opened and closed positions. The ECU **144** may then send in response to such a determination the appropriate actuation signals **144.sub.c** (by cutting the power flow to the motor **142** for example) resulting in the motor **142** to stop to allow a manual override/control of the closure panel **14** by a user. Conversely, and as an example of an object or obstacle detection functionality, when the ECU **144** is in a power open or power close mode and the Hall-effect sensors **148.sub.a** indicate that a speed of the motor **142** is less than a threshold speed (e.g., zero) and a current spike is detected (in the case where ripple counting is employed to determine the operation of the motor **142**), the ECU **144** may determine that an obstacle or object is in the way of the closure panel **14**, in which case the ECU **144** may take any suitable action, such as sending an actuation signal **144.sub.c** to turn off the motor **142**, or sending an actuation signal **144.sub.c** to reverse the motor **142**. As such, the controller ECU **144** receives feedback from the Hall-effect sensors **148.sub.a**, or from a current sensor (not shown) and renders control decisions locally to biasing member **37** to ensure that a contact or impact with the obstacle and the closure panel **14** has not occurred during movement of the closure panel **14** from the closed

position to the opened position, or vice versa. An anti-pinch functionality may also be performed in a similar manner to the obstacle detection functionality, to particularly detect an obstacle such as a limb or finger is present between the closure panel **14** and the vehicle body **11** about the nearly fully closed position during the closure panel **14** transition towards the fully closed position.

(59) While the ECU **144** has been described as executing decisions and algorithms to operate the motor **142** for obstacle detection, automatic close and open, anti-pinch functionality, the controller ECU **144** may also be programmed to perform other functions. For example, such functionality executed by the ECU **144** may include dynamic closure panel speed and torque control adjustments (i.e. the ECU **144** can detect an unlevel vehicle surface **9** and operate the motor **142** to adjust the torque output of the motor **142** required to move the closure panel **14**), stop-and-hold position request (e.g. the ECU **144** can detect a user manually stopping the closure panel **14** and maintain the closure panel at this position), requests for motion (e.g. the ECU **144** can detect a user manually pushing the closure panel **14** and operate the motor **142** to move the closure panel **14** to the fully closed position, and the ECU **144** can detect a user manually pulling the closure panel **14** to move the closure panel **14** to the fully opened position), falling gate detection (e.g. the ECU **144** can detect a movement of the closure panel **14**, due to for example a failed counterbalancing spring or biasing device and apply a brake mechanism (not shown) provided in the biasing member **37** to hinder the movement of the closure panel **14** towards the fully closed position), time based detected obstacle (e.g. the ECU **144** can track the length of time by implementing a counter function to determine the period of time over which the ECU **144** has determined that the speed of the motor **142** is less than a threshold speed (e.g., zero) and a current spike is detected (in the case where ripple counting is employed to determine the operation of the motor **142**) indicative of an obstacle, zone detected obstacle (e.g. the ECU **144** can detect the presence of an obstacle in the path of movement of the closure panel **14** in the configuration where the ECU **144** is in communication with an proximity sensor, such as a radar sensor, is provided either externally to the biasing member **37** such as in latch **43**, or internally in the biasing member **37**. In the configuration where the radar sensor is provided internally the biasing member **37**, the radar sensor may be provided on the PCB **145** and aligned with an aperture or port in the housing **35**, if the housing **35** is manufacture from metallic material, to allow for radar signals to be transmitted and received therefrom. The ECU **144** in such a configuration may be configured to process the radar signal (i.e. FMCW or Doppler signals) and determine if obstacles are present), current detect obstacle (e.g. ECU **144** can determine a current spike is detected over the signal lines from the motor **142** to the ECU **144** in the case where ripple counting is employed to determine the operation of the motor **142**. It is recognized that ripple counting techniques may do away with the Hall sensors **148.sub.a** and magnet **151** and be replaced with processing by the ECU **144** involving calculating the ripple frequency using sensed the motor current measurements that has been bandpass filtered to process the relevant spectrum of the motor current), full open position request (e.g. the ECU **144** may automatically move the closure panel **14** to a fully opened positioned), learn completed request (e.g. the ECU **144** can detect the operating characteristics of the closure panel **14**, such as the torque profile, and create an opening profile stored in the memory **144.sub.b** after operating the closure panel **14** in a training mode and sensing the movement represented by the feedback received by the sensors **148.sub.a**, after which the ECU **144** can optimize the torque output and speed for example of the motor **142** for moving the closure panel **14** after having learned the movement characteristics of the closure panel **14** during the training mode operation of the biasing member **37**, for example immediately after the biasing member **37** having been installed on the motor vehicle **10**, or after an aftermarket accessory has been added to the closure panel **14** resulting in an increase its weight and changing its moving characteristics), motor motion (e.g. the ECU **144** can operate the motor **142** at different torques and speeds depending on the position of the closure panel **14**), adjustable stop position (e.g. the ECU **144** can operate the motor **142** to hold the closure panel **14** at a certain position based upon a received command signal **144.sub.e** which may include

position and angle data), short to ground (e.g. the ECU **144** can diagnose operational faults in the control circuitry or electric motor **142** and communicate to the external control system the fault, maintenance and troubleshooting codes and the like), short to battery (e.g. the ECU **144** can diagnose power supply line faults i.e. faults in Electrical power signal line **106** or motor blade connections), open circuit for all components like (e.g. the ECU **144** can diagnose operational faults in the control components such as the motor **142**, the Hall effects sensors **148.sub.a**, transient suppression faults, etc.). Other functions may include an unpowered rapid motor motion function which may also be performed by the ECU **144** (e.g. the ECU **144** may electrically disengage an electromechanical clutch of the biasing member **37**). Other operating functions as determined and executed by the ECU **144** are also contemplated and are not limited to those described herein.

(60) In view of the above-described configuration, it is recognized that the ECU **144** can have its own software (e.g. set of operating instructions **159**) stored in a physical memory **144.sub.b** for example to configure operation of the motor **142** and optional receipt of the external signals (e.g. command signals **144.sub.e**) via the electric cabling **110** to the controller ECU **144** (e.g. from the handles, latch **43**, and/or other vehicle control system **16**) and transmission of the signals (e.g. feedback signals **144.sub.h**) externally based on the detection by the internal electronic sensors (for example Hall-Effect sensors **148.sub.a**, ripple counting sensing) via the communication interface **144.sub.d**.

(61) Now referring to FIG. **15**, in addition to FIG. **12**, there is provided is a flow chart illustrating a method of operation **200** of the electronic motor assembly **15** as controlled by ECU **144** according to instructions representing steps of the method **200** as stored in memory **144.sub.b** and executed by the processor **144.sub.a**, in accordance with an example embodiment. A person of ordinary skill in the art will understand that the ECU **144** of the electronic motor assembly **15** may be programmed to enable the practice of the method **200**. Moreover, program instructions for use with the ECU **144**, stored on memory **144.sub.b** or other similar computer readable medium or computer program product including the instructions recorded thereon, may direct the ECU **144** to facilitate the practice of the method **200**. In particular, the sequences of instructions which when executed cause the method described herein to be performed by the ECU **144** may be contained in the memory **144.sub.b** in electrical communication with the ECU **144**. This data carrier product may be loaded into and run by the ECU **144**. In addition, the sequences of instructions which when executed cause the method described herein to be performed by the ECU **144** may be contained in a computer software product or computer program product (e.g., comprising a non-transitory medium) stored in the memory **144.sub.b** according to one embodiment of the invention. This computer software product or computer program product may be loaded into and run by the ECU **144**. Moreover, the sequences of instructions which when executed cause the method **200** described herein to be performed by the ECU **144** may be contained in an integrated circuit product integrated within the processor **144.sub.a** (e.g., a software module **500** or hardware modules **502** which may include a coprocessor or memory according to one embodiment). This integrated circuit product may be installed in the ECU **144**, as illustrated in FIG. **12**. Other methods of operation of the electronic motor assembly **15** as controlled by ECU **144** may be achieved in a similar manner.

(62) Still referring to FIGS. **12** and **15**, shown is an operational flowchart **200** example of the electromechanical biasing member **37** of FIG. **4**. At step **202**, the ECU **144** receives a command signal **144.sub.e** communicated over the electrical connection(s) **110**. At step **204**, the ECU **144** processes via the electrical components **148**) the command signal **144.sub.e** (as received by the communication interface **144.sub.d** coupled to the electrical connections **110** and the processor **144.sub.a**) to produce the actuation signal **144.sub.r** for delivery to the motor **142**. At step **206**, the ECU **144** sends the actuation signal **144.sub.c** to the motor **142**, i.e. positioned adjacent to the ECU **144** as shown in FIG. **5**. It is recognized that the motor **142** can also draw operational power from the electrical connection(s) **110** (either directly and/or via the electrical components **148** of the ECU **144** e.g. processor **144.sub.a**) in order to process the actuation signal **144.sub.c** and rotate the

motor shaft 157. At step 208, the ECU 144 receives one or more sensor signals 144.sub.f from one or more sensors (e.g. hall sensors 148.sub.a), processes the sensor signals 144.sub.f via the other electrical components 148 and sends the results of the sensor signal processing (e.g. indicating the positional state of the closure panel 14) to the control system 16 via the electrical connection(s) 110. For example, the sensor signals 144.sub.f represented as analog voltage signal as generated by the sensors 148.sub.a may be processed by the ECU 144 and the result of such processing (i.e. filtering, analyzing) is digitized by the communication interface 144.sub.d for transmission as a digital message 144.sub.h depicted by an arrow to a remote master vehicle system controller 16 for processing thereby. It is recognized that the action of control system 16 in response to receiving the digital message 144.sub.h can be simply a display by the vehicle systems for informing an occupant of the vehicle 11 (via the results 144.sub.h), such as a closure panel 14 active open alert, closure panel 14 fully closed or fully open alert, closure panel 14 obstacle detection alert, and/or control system 16 can contain control processing logic for implementing further process control of the results 144.sub.h, for example the vehicle system may in conjunction with communication with other vehicle systems, such as a drive or park state module of the vehicle, process the results 144.sub.h in conjunction with the drive state to issue a “CLOSE” command to the ECU 144 after the results 144.sub.h indicates that the closure panel 14 is in the fully opened position, but the vehicle 10 has now been transitioned from a parked state to a drive state. It is recognized that the ECU 144 can receive a subsequent command signal 144.sub.e from the control system 16 as further action when the remote control system 16 is operating as a master controller, and where the ECU 144 is operating as a slave to the master controller. Alternatively, the ECU 144 can generate its own further actuation signal 144.sub.c based on the results 144, of the sensor signal processing, an example method of operation of which is now provided.

(63) As illustrated in FIG. 16, a method of operating or moving a closure panel (e.g. lift gate 14) using the biasing members 37 incorporating the electronic motor assembly 15 is now described in more detail. In accordance with an illustrative embodiment, the closure panel 14 operation, or moving algorithm, disclosed herein is described for a command for transitioning the closure panel 14 from a fully closed to a fully open position when the ECU 144 of biasing member 37.sub.1 is operating as a master controller having received a generic open command from an external vehicle system, and optionally if the second biasing member 37.sub.2 shown in FIG. 2 is operating as a slave to the master controller ECU 144 of biasing member 37.sub.1. Other commands for operating the closure panel 14 may be provided, such as but not to a command for transitioning the closure panel 14 from the fully opened to the fully closed position, a command for transitioning the closure panel 14 from the fully closed to a partially open position, a command for stopping the closure panel 14 at a third position hold during the transition of the closure panel 14 between the fully opened and the fully closed position. It is understood that the ECU 144 of the electronic motor assembly 15 may be programmed to enable the practice of the method 400, in a like manner as described herein above for the practice of the method 200.

(64) Still referring to FIG. 16, the method 400 includes the step of 402 receiving a command signal 144.sub.e (e.g. a generic high level “Fully Open” command to move the closure panel 14 to a fully open position) from an external system (e.g. from Body Control Module in operable communication with the biasing member 37 having received a closure panel open signal from a wireless key fob remote from the vehicle 10 as but an example, or from a switch directly in operable communication with the biasing member 37) via the communication interface 144.sub.d. Then the method 400 continues by step 404 processing the command signal 144.sub.e by the internal control unit (e.g. ECU 144). The step of 404 processing the command signal 144.sub.e may include decoding an encoded digital message transmitted over the electrical connection(s) 110 in accordance with a communication protocol, for example a message encoded in accordance with the LIN protocol as discussed hereinabove, to extract/decode the command signal 144.sub.e. In such a manner, the communication with the biasing member 37 is simplified since the vehicle system 16

issuing the closure panel command to the biasing member 37 does not have to provide any additional control information or understand the operating characteristics or lower hardware level specific operating procedures of the biasing member 37, for example which may be required during the execution of the command as provided by supplementary electrical communication lines carrying feedback signals from the electric motor 142, or other sensors 148.sub.a, such as hall sensors, as well as additional functionality and instructions which the vehicle system 16 must be so equipped with requiring additional software and/or hardware and added complexity. Additionally, the vehicle control system 16 may issue a single operating command to multiple biasing members 37 interconnected over a network for moving the closure panel 14 in synchronicity, for example as shown in FIG. 2 whereby two biasing members 37.sub.1,2 may operate in tandem to move the closure panel 14. Local decision control and operation of the motor 142 for moving the closure panel 14 is based upon the processing by each ECU 144 of the two biasing members 37.sub.1,2 separately, or one ECU 144 for example the first biasing member 37.sub.1 may operate as the master controller issuing commands and receiving feedback from the second biasing member 37.sub.2. In the former configuration, the control of each of the two biasing members 37.sub.1,2 is local and internal to each, each capable of operating in tandem based on the sensed signals of the motor operation 142 indicative of the other one of the two biasing members 37.sub.1,2 actions on the closure panel 14. Additionally, wiring/cabling required for feedback sensors 148.sub.a to the remote vehicle control system 16 is eliminated and/or reduced, and sensitivity to electromagnetic noise coupled to such feedback lines external the biasing members 37 is also reduced since the feedback circuits, lines and sensors are housed within the biasing members 37 and isolated from such external interference. The method 400 proceeds with the step of 406 generating an actuation signal 144.sub.c, for example a Pulse Width Modulated voltage signal as generated by the FETs 148.sub.b as controlled by the ECU 144 in a manner as described hereinabove, with the output of the FETs 148.sub.b to be supplied to the electrical motor 142, such as a brushed motor 142 as an example, or a ECU 144 may generate a field vector control signal to control the motor driver 144.sub.g to supply a three-phase PWM motor voltage output derived from vector signal for the motor 142 in the configuration where the motor 142 is a brushless motor 142. The generated actuation signal 144.sub.c may be based on the desired operating sequence of the closure panel 14 based on the command signal 144.sub.e, either in accordance with a predetermined sequence as stored in memory 144.sub.b and/or in combination with actively determined of the state of the closure panel 14 operating parameter, as sensed by the feedback sensors 148.sub.a; and other desired closure panel movement profiles.

(65) The method 400 continues further with the step of 410 sending the actuation signal 144.sub.c to the electric motor 142, for example by controlling the FETS 148.sub.b provided for on the PCB 145 which are electrically coupled to the motor blade receptacles 147.sub.a via the connection with the motor blades 147.sub.b as described hereinabove, to control operation of the electric motor 142. In the configuration where the biasing member 37.sub.1 operates as the master controller, and the other biasing member 37.sub.2 operates as the slave controller, the step of 410 may also include sending a command to the ECU 144 of the other biasing member 37.sub.2 to generate locally the actuation signal 144.sub.c to be supplied to the electric motor 142 of the other biasing member 37.sub.2. The actuation signal 144.sub.c is generated by the local FETs 148.sub.b as controlled by the ECU 144 in a manner as described hereinabove. Localizing the FETs 148.sub.b near or proximate the motor blade receptacles 147.sub.a provides an inherent heat sink for the FETS 148.sub.b to dissipate heat generated by the FETS 148.sub.b during operation. Next, the method 400 proceeds with the step of 412 receiving feedback from the sensors 148.sub.a indicative of the operating status of the motor 142 or the closure panel 14, or other operating parameters, such as sensing the speed of rotation the electrical motor shaft 157 from which the closure panel 14 movement, such that speed of the closure panel 14 can be inferred as calculated by the ECU 144, or determining the position of the closure panel 14 as determined by tracking the electric motor shaft

157 revolutions, or other position detection techniques, or detecting if an obstacle is present in the moving path of the closure panel **14** as determined for example by the interpretation of differences in electrical motor shaft revolutions as detected by the hall sensors **148.sub.a** and an expected motor speed based on the inputted actuation signal **144.sub.c**, or by sensing of an obstacle by a proximity sensor in communication with the ECU **144**; or receiving feedback related to the angle of inclination of the vehicle **10** relative to a unlevel vehicle surface **9** to perform motor torque output adjustments to the actuation signals **144.sub.c**. Such receiving feedback from the sensors **148.sub.a** can be done either locally by the master controller ECU **144** for determining the operating status of the motor **142** of the biasing member **37.sub.1** when operating as the master controller, or remotely as received by the master controller ECU **144** of biasing member **37.sub.1** as transmitted over communication network from the slave controller ECU **144** of the other biasing member **37.sub.2**. (66) Next the method **400** continues with the step of **414** determining if the command has been met, for example by comparing stored closure panel state information stored in memory **144.sub.b** and comparing with the sensor data received by the ECU **144** during operation of the closure panel **14**.

(67) Next the method **400** continues with the step of **416** sending the actuation signal **144.sub.c** to stop the electric motor **142** upon the command being met e.g. the electrical motor **142** will be stopped upon the closure panel **14** reaching a desired position e.g. fully open. A similar command signal (e.g. subcommand signal **1441**) transmitted over the communication network by the ECU **144** of the master controller of the biasing member **37.sub.1** to the slave controller ECU **144** of the other biasing member **37.sub.2** may also be performed. The method **400** further continues with the step of **416** returning to the step of **410** generating an actuation signal **144.sub.c**, which may be adjusted by the ECU **144** as a result of the feedback by the sensors **148.sub.a**, to the electric motor **142**. if the command condition has not be met e.g. if the closure panel **14** has not moved to a fully opened position. If at step **414** it is determined that the command condition has been met, then the next step of the method **400** is 416 sending the actuation signal **144.sub.c** to stop the electric motor **142**.

(68) Now referring to FIG. **17A**, there is illustrated a distributed master/slave architecture system **599** for operating a closure panel **14** using a biasing member **37.sub.1** operating as the master controller **602** for receiving generic commands (e.g. **144.sub.e**) from a remote vehicle control system **16**, and the other biasing member **37.sub.2** operates as the slave controller **608**, for receiving command signals (e.g. subcommand signals **1441**) from the master controller **602** and transmitting feedback signals **144.sub.h** to the master controller **602**, with the signals being transmitted and received over the communications link **590**, such as a vehicle communication network (e.g. using at least in part electrical connection(s) **110**). In accordance with this illustrative embodiment the master controller **602** controls its local motor **142**, as well as the remote motor **142** of the slave biasing member **37.sub.2** to move the closure panel **14**. In an illustrative embodiment, the remote vehicle control system **16** is a latch **43**, such as a smart latch having local intelligence and processing capabilities. In another illustrative embodiment, the remote vehicle control system **16** is a latch controller **116** provided separate from the latch **43**.

(69) Now referring to FIG. **18A**, in addition to FIG. **17A**, there is provided a schematic diagram exemplifying the exchange of communications of command signals to move a closure panel **14** from a first position to a second position, with the exchange of communications occurring between a remote vehicle control system **16** issuing a generic high level open command signal **600**, the ECU **144** of a master controller **602** of a first biasing member **37.sub.1** generating a local actuation signal **604** to operate a local motor **142** and issuing a second subcommand signal **606** (e.g. **1441**) for transmission over a communication network (e.g. communication link **590**) to the ECU **144** of a slave controller **608** of a second biasing member **37.sub.2**. The ECU **144** of a master controller **602** of a first biasing member **37.sub.1** also receiving a feedback signal **611** (e.g. **144.sub.t**) of the operation of the local motor **142**. The ECU **144** of a slave controller **608** of a second biasing

member **37.sub.2** generates a local actuation signal **610** (e.g. **144.sub.c**) to operate a local motor **142** and generates a feedback signal **612** (e.g. **144.sub.h**) of the operation of the local motor **142** and transmits the feedback signal **612** over the communication network **590** to the ECU **144** of a master controller **602** of a first biasing member **37.sub.1**. The ECU **144** of a master controller **602** of a first biasing member **37.sub.1** processes the feedback signals **611**, **612**. In the illustrative embodiment, the command has not been fulfilled at this point and the ECU **144** of a master controller **602** of a first biasing member **37.sub.1** generating in response another local actuation signal **604** to operate a local motor **142** and issuing another subcommand signal **606** for transmission over the communication network **590** to the ECU **144** of a slave controller **608** of a second biasing member **37.sub.2**. The ECU **144** of a master controller **602** of a first biasing member **37.sub.1** also receiving a feedback signal **611** of the operation of the local motor **142** and the ECU **144** of a slave controller **608** of a second biasing member **37.sub.2** generates another local actuation signal **610** to operate a local motor **142** and generates another feedback signal **612** of the operation of the local motor **142** and transmits the additionally sensed feedback signal **612** over the communication network **590** to the ECU **144** of a master controller **602** of a first biasing member **37.sub.1**. The ECU **144** of a master controller **602** of a first biasing member **37.sub.1** processes again the feedback signals **611**, **612**. In the illustrative embodiment, the ECU determines the command has been fulfilled (e.g. closure panel **14** has been moved to the desired position communicated in the generic command signal **600**, and ECU **144** has compared the actual detected position of the closure panel **14** and compared the position to an expected position representative of a fulfilled command signal as an illustrative example) and the ECU **144** of a master controller **602** of a first biasing member **37.sub.1** generating another local actuation signal **604** to stop the local master motor **142** and issuing another subcommand signal **606** for transmission over the communication network **590** to the ECU **144** of a slave controller **608** of a second biasing member **37.sub.2** to stop the local slave motor **142** of the second biasing member **37.sub.2**. A command complete signal **616** (e.g. **144.sub.h**) is transmitted to the remote vehicle control system **16** to perform a vehicle system action, such as activate an alert (visual, or audio) or perform other actions as desired by the remote vehicle system **16**.

(70) Now referring to FIG. **17B**, there is illustrated a distributed master/slave architecture system **598** for operating a closure panel **14** using a biasing member **37.sub.1** operating as the slave controller **620** for receiving detailed operating commands (e.g. **144.sub.e**) from a remote vehicle master controller system **16**, and the other biasing member **37.sub.2** operating as another slave controller **622**, for also receiving detailed operating commands from a remote vehicle master controller system **16**. Both slave controllers **620**, **622** are configured, in a manner as described hereinabove to transmit feedback signals to the master controller **618**, with the signals being transmitted and received over the communications link **590**, such as a vehicle communication network. In accordance with this illustrative embodiment the master controller **16** controls the local motors **142** of each biasing member **37.sub.1**, **37.sub.2** to move the closure panel **14**. In an illustrative embodiment, the remote vehicle master controller system **16** is a latch **43**, such as a smart latch having local intelligence and processing capabilities. In another illustrative embodiment, the remote vehicle master controller **16** is a latch controller **116** provided separate from the latch **43**.

(71) Now referring to FIG. **18B**, in addition to FIG. **17B**, there is provided a schematic diagram exemplifying the exchange of communications of command signals to move a closure panel **14** from a first position to a second position, with the exchange of communications occurring between a remote vehicle master control system **16**, generically referred to using the reference number **618**, issuing a detailed level open command signal **630**, the ECU **144** of a slave controller **622** of a first biasing member **37.sub.1** generating a local actuation signal **623** to operate a local motor **142** in response to receiving the command signal **630**, and remote vehicle master control system **16**, **618** issuing a detailed level open command signal **630** for transmission over a communication network

590 to the ECU 144 of a slave controller 622 of a second biasing member 37.sub.2, the ECU 144 of a slave controller 622 of the second biasing member 37.sub.2 generating a local actuation signal 624 to operate a local motor 142 in response to receiving the command signal 630. The level of detail of the open command signal 630 may vary depending on the system configuration and may include lower level details such as motor speed, position, control actions, operating modes, adjusted commands based on received feedback from each slave controller 620, 622 etc. The ECU 144 of a slave controller 620 of a first biasing member 37.sub.1 also providing a feedback signal 633 (e.g. 144.sub.h) of the operation of the local motor 142. The ECU 144 of a slave controller 622 of a second biasing member 37.sub.2 generates a local actuation signal 624 to operate a local motor 142 and generates a feedback signal 631 (e.g. 144.sub.h) based on the operation of the local motor 142 and transmits the (e.g. 144.sub.h) over the communication network 590 to the remote vehicle master controller system 16. The ECU 144 of the slave controller 620 of a first biasing member 37.sub.1 processes internally sensed feedback signals 633. The ECU 144 of the slave controller 620 of the second biasing member 37.sub.2 processes internally sensed feedback signals 631. In the illustrative embodiment, vehicle master controller system 16, 618 determines that command has not been fulfilled based on the received feedback signals 631, 633 and the vehicle master controller system 16, 618 issues another detailed level command signal 630 to the slave controllers 620, 622. In response, ECUs 144 of a slave controllers 620, 622 of a first biasing member 37.sub.1 and second biasing member 37.sub.2 generate another local actuation signal 623, 624 to operate a local motor 142. The ECU 144 of a slave controllers 620, 622 of a first biasing member 37.sub.1 and second biasing member 37.sub.2 also sense and transmit to remote vehicle master controller system 16, 618 newly acquired feedback signals 631, 633 of the operation of the local motor 142 in response to the new local actuation signals 623, 624 over the communication network 590 to the vehicle master controller system 16. The vehicle master controller system 16, 618 processes again the new feedback signals 631, 633 to determine if the command condition has been met. In the illustrative embodiment, the command has been fulfilled (e.g. closure panel 14 has been moved to the desired position) and the vehicle master controller system 16, 618 issuing another command signal 640 for transmission over the communication network 590 to the ECUs 144 of a slave controller 620, 622 of first biasing member 37.sub.1 and second biasing member 37.sub.2 to stop the local motors 142. ECUs 144 of a slave controllers 620, 622 of the first biasing member 37.sub.1 and second biasing member 37.sub.1 generating in response to command signal 640 another local actuation signal 626 to stop the local motors 142. It is recognized that during operation of motors 142 by the slave controllers 620, 622, the slave controller 620, 622 may operate in modes, either as directed by the vehicle master controller system 16, 618, or as automatically performed. For example, the slave controller 620, 622 may be set to operate in an obstacle detection mode whereby despite the issued command signals 630 to move the closure panel, in the even the slave controller 620, 622 locally determine an obstacle is present, the slave controller 620, 622 will override the command signal and issue an actuation signal in response to the detected obstacles e.g. stop the motor 142.

(72) Referring to FIGS. 17A, 17B, 18A, 18B, the master controller (e.g. master controller 602, control system 16) can be a latch controller (i.e. a controller on-board of the latch 43 represented as an e-latch). Alternatively, the master controller (e.g. master controller 602, control system 16) can be represented by latch control system 116 (see FIG. 19). In terms of communications of signals (e.g. 144.sub.e, f, h) between the master controller (e.g. 602, 618 16) and the slaver controller(s) (e.g. 620, 622) the ECU 144 (of the slave controller 620, 622) receives the command signal (e.g. 144.sub.e) communicated over the electrical connection(s) 110 as received from the master controller (e.g. 602, 618, 16). The ECU 144 processes (via the electrical components 148) the command signal 144.sub.e to produce and send the actuation signal 144.sub.c for local delivery to the motor 142 for processing by the motor 142. The motor 142 can draw operational power from the electrical connection(s) 110 in order to process the actuation signal 144.sub.c and rotate the

motor shaft **157**. The ECU **144** can also receive one or more sensor signals **144**, from one or more sensors, process the sensor signals **144.sub.f** via the other electrical components **148** and send the results of the sensor signal processing to the master controller **602,618,16** via the electrical connection(s) **110**. Alternatively, the sensor signals **144.sub.f** can be sent (unprocessed or partially processed) by the ECU **144** to the remote master vehicle system controller **602, 618, 16** for processing thereby.

(73) The ECU **144** can receive a subsequent command signal **144.sub.e** (e.g. stopping the motor **142**) from the master controller (e.g. **16, 602, 618**) as further action and where the ECU **144** is operating as a slave to the master controller. Alternatively, the ECU **144** can generate its own further actuation signal **144.sub.c** (e.g. stopping the motor **142**) based on the results **144**, of the sensor **148.sub.a** signal processing.

(74) The received feedback from the sensors **148.sub.a** can be indicative of the operating status of the motor **142** or the closure panel **14**. Other operating parameters as the received feedback can be parameters such as but not limited to any one or more of: sensing the speed of rotation the electrical motor shaft **157**; determining the position of the closure panel **14**; detecting if an obstacle is present in the moving path of the closure panel **14**; and/or receiving feedback related to the angle of inclination of the vehicle **10** relative to a unlevel vehicle surface **9**. Such receiving feedback from the sensors **148.sub.a** can be done locally by the ECU **144** of the slave controller ECU **144**. Alternatively, receiving feedback from the sensors **148.sub.a** can be done remotely as received by the master controller ECU **144** as transmitted over communication network from the slave controller ECU **144**.

(75) In operation, the control system **16** controls the operation of an electromechanical strut **37** for moving a closure panel **14** between an open position and a closed position relative to a motor vehicle body **11**, the electromechanical strut comprising a housing **35** comprising an inner cavity **124**, and an electric motor **142** positioned within the inner cavity **124**, the control system comprising: a module **116** associated with a vehicle latch **34**, the module comprising a master controller **602** configured to issue commands **144.sub.e** to move the closure panel **14** between an open position and a closed position; and a slave controller **620** remote from and in communication with the master controller **602** to receive the commands, the slave controller **620** comprising: a control unit **144** positioned within the inner cavity and configured to drive the electric motor **142**, the controller comprising: at least one controller board **145** juxtaposed to the electric motor **142**; at least one computing device mounted to the at least one controller board and configured to execute instructions to generate a control command **144.sub.e** to drive the electric motor **142**, the at least one computing device comprising at least a data processor **144.sub.a** and a memory **144.sub.b** storing the instructions **159**, the memory **144.sub.b** being coupled to the data processor **144.sub.a**; and a plurality of control components **148** mounted to the at least one controller board **145** configured to supply a drive signal **144.sub.c** based on the control command **144.sub.e** to the electric motor and receive signals **144.sub.h** representative of an electric motor operation.

(76) In the above illustrative method and systems, the processing of a received command signal **144.sub.e** from an external vehicle control system **16** is performed local and internal to the biasing member **37**, as well as the processing of feedback signal (e.g. **148.sub.a**) from the biasing member **37**, and the generation of control signals (e.g. actuation signals **144.sub.c**) to be supplied to the electric motor **142**, alleviating the control complexity required by the external vehicle system **16**, simplifying communication with the control system **16** and one or more biasing members **37**, as well as reducing wiring and installation complexity, as will now be explained in more details.

(77) Now referring to FIG. **19** to FIG. **23**, there is illustrated an example of the latch control system **116** providing the control signals **108** to the integrated electronic motor assembly **15** of a biasing member **37** forming part of a distributed biasing member control system **1000**, in accordance with an illustrative embodiment. The system is illustratively a remotely located control system to the housing **35** of the biasing member **37**, such as a latch control system **116** located within the closure

panel **14** as illustrated in an embodiment of FIG. 2. Latch control system **116** may optionally also be located on the vehicle body **11**. Latch control system **116** includes a housing **118** for enclosing various latch control electronic components **117**. The housing **118** may include mounting features, such as apertures **120** for fixing the housing **118** to the vehicle body **11** or within closure panel **14** such as a door module (not shown) using screws or other connectors e.g. snaps, rivets, nuts and bolts, or otherwise. The latch control system **116** includes a printed circuit board **123** enclosed by the housing **118** for supporting the various latch control electronics **117** thereto in coordinated electrical communication with one another. Such electronic components **117** may include hardware and software components such as microcontrollers **125** and memory modules **127**, such as microchips, for storing instructions for execution by the microcontrollers **125**, as well as resistors, inductors, and capacitors and other signal conditioning/supporting components for operating the microcontroller **125** and memory modules **127** of the latch control system **116**. For example, instructions stored on the memory module **127** may be related to various system modules **131** as illustrated in FIG. 21, for example application programming interfaces (API) modules **131.sub.1** for a latch API, drive API, digital input output API, Diagnostic API, Communication API, and communication drivers for LIN communications and CAN bus communications. Also provided are microcontroller drivers **131.sub.2** such as LIN drivers, CAN drivers, SPI, I/O control, power control, memory manager and mode control for microcontroller management, control functions, and input/output control modules. Also provided are mechanical subsystem modules **131.sub.3**, such as for latch control of latch **43** and interfacing therewith. Other software and/or hardware architecture modules **131** may also be provided as part of the illustrative distributed system architecture **1000** of FIG. 21. While modules are described as being loaded into a memory, such as memory module **127**, it is understood that the modules could be implemented in hardware and/or software. Absent from the latch control system **116** forming part of a distributed biasing member control system **1000**, in accordance with an illustrative embodiment are biasing member **37** related control modules, such as FET hardware and software **131.sub.4**, **170**, FET APIs **131.sub.5**, **190**, biasing member diagnostics control **131.sub.6**, **139**, timers **131.sub.7** and sensor related modules and systems **131.sub.8**, as but examples, which can now be localized to the integrated electronic motor assembly **15**, as illustrated in FIG. 20.

(78) As illustrated in FIG. 22, and with further reference to FIGS. 20 and 21, the latch control system **116** operation and physical configuration is simplified when used in combination with the integrated electronic motor assembly **15** to control the biasing member **37** as described herein above. Due to the integrated controller ECU **144** provided within the integrated electronic motor assembly **15**, control and operation decisions can be made/calculated locally to the biasing member **37** as compared to currently known latch control systems (indicated by reference number **116.sub.before** in ghosted view of FIG. 23) where control and operation decisions for the biasing member **37** would have been made/calculated remotely at the latch control system **116.sub.before**. It is recognized that various hardware and software components and modules typically associated with the existing latch control system **116.sub.before** can be consequently reduced or eliminated from the latch control system **116**, and rather provided within the biasing member **37** with integrated electronic motor assembly **15** as described herein above. For example, mechanical subsystem control modules **130** functionality (i.e. related hardware and software components), such as the biasing member' and drive control modules **132**, as well as the FET drivers **170**, FET API modules **190** (e.g. **131.sub.5**), the fault manager module **136**, a Pulse Width Modulation driver **98** and analog sensor modules **138** for feedback sensors **148.sub.a** can now reside locally to the integrated electronic motor assembly **15**, as opposed to residing with the latch control system **116**. It is recognized that the component count of the latch control system **116.sub.before** can be reduced, as illustrated by the ghosted portion of the components eliminated in FIG. 20 and in FIG. 23, and the footprint of the latch control system **116.sub.before** e.g. the PCB size and physical dimension of the housing **118**, can also be reduced, as also shown in ghosted illustration in FIG. 23,

when using the latch control system **116** in conjunction with the integrated electronic motor assembly **15** as described above. For example, as illustrated in phantom outline in FIGS. **20** and **23**, the FET drivers **170** and corresponding PCB footprint **180** in latch control system **116.sub.before** required to mount the FETs **170** thereto can be eliminated having been integrated as integrated FETs **170** (i.e. FETS **148.sub.b**) into the integrated electronic motor assembly **15**, i.e. the FETs **170** are mounted to the PCB **145** of the integrated electronic motor assembly **15**, thereby reducing the packaging size of the latch control system **116.sub.before**. Also, corresponding FET control software modules, such as the FET API **131.sub.5**, shown as block **190** in FIG. **22**, can be moved from the latch control system memory module **127** (of the latch control system **116.sub.before**) and moved locally to the memory **144.sub.b** of the integrated electronic motor assembly **15**, and further having FET control software instructions executed locally by the processor **144.sub.a**. Also, wiring/cabling carrying electrical signals from the FETs **170** previously provided on the latch control system **116.sub.before** can also be removed, reducing wire count between the latch control system **116** and the integrated electronic motor assembly **15**, being replace with the electrical connection(s) **110**. In lieu, a simplified high level generic command instruction can be sent over the electrical connection(s) (e.g. communication lines) **110** by the latch control system **116**, for example a command either for activating the opening of the closure panel **14**, closing the closure panel **14**, can be transmitted to the integrated electronic motor assembly **15** with the integrated electronic motor assembly **15** handling and processing the lower lever hardware specific operations to execute and fulfill high level command, as discussed hereinabove. When reference is made herein to an Application Programming Interface (API), it may also be understood to include the underlying related software and/or hardware of that labelled API i.e. the FET API may include the hardware and/or software required for controlling the FETs through which the FET API may interface and communication with.

(79) While the control and hardware components related to the FETs, such as the FET drivers **170**, and FET API modules **190** have been illustrated in FIG. **23** as now integrated with the integrated electronic motor assembly **15**, it is understood that other control and hardware components can be integrated with the integrated electronic motor assembly **15** as would have been implemented by the remote latch control system **116.sub.before** e.g. hall effect drivers **193**, Diagnostic API **139**. Specific calculations for the control of the biasing member **37** i.e. speed, obstacle detection, can be carried out locally to the biasing member **37** in a manner as described above. Also the distributed and now local control provided by the ECU **144** of a biasing member **37** can provide for an easily scalable system based on the master and slave architecture, whereby the latch control system **116** operates as the master controller, while the ECU(s) **144** of the biasing member(s) **37**, which may include more than one biasing members **37.sub.1,2,3** as shown in FIGS. **17B** and **23** in communication with the master latch control system **116**, thereby operating as slaves to the latch control system **116** acting/operating as the master controller. As such only a simplified generic commands, such as STOP or START commands can be issued by the latch control system **116** acting as the master controller to the one or more (e.g. multiple) slave controller(s) (i.e. one or more respective ECUs **144**) without the necessity of the latch control system **116** having to consider the number of slaves (i.e. number of ECUs **144**), process the feedback signals from the slaves, require hardware/software to operate specific functionality of the biasing member **37** (e.g. if biasing members are of different types), or process the control signals required to operate each integrated electronic motor assembly **15**, which is now undertaken by the respective ECU **144** contained within the housing **35** of the respective biasing member **37**.

(80) As such, the above described electromechanical biasing member **37** with integrated electronic motor assembly **15** provides for an integrated controller ECU **144** with other electronic components **148** (e.g. sensors, drivers, SIP electronics, controller, etc.) all housed within a motor housing (e.g. the lower housing **112** portion) of the housing **35**. It is recognized that having the components **148** in close proximity to the motor **142**, i.e. within the housing **35**, facilitates the reduction of signal

noise that would otherwise be present when such unprocessed (either in whole or in part) signals are transferred external to the housing **35** to a remotely located processor of a vehicle control system **16** (i.e. using the electrical connection(s) **110** going through a wall of the housing **35**). It is also recognized that positioning of the ECU **144** and the motor **142** within the housing **35** provides for a more compact footprint of the electromechanical biasing member **37**, such that the internally controlled electromechanical biasing member **37** has greater applicability in tighter or industry standard footprint requirement applications, and can replace existing biasing members of like dimensions without need for modification of the receiving footprint. As such, the stacked arrangement of both the ECU **144** (e.g. the PCBs **145**) and the motor **142** along the longitudinal axis **41** of the electromechanical biasing member **37** provides for the ability of the integrated biasing member **37** to be installed in existing configurations without having to substantively modify the existing mounting area in the vehicle **10** to accommodate the electromechanical biasing member **37**. In terms of having the sensor components **148** located adjacent to the signal processing components **148** of the same or adjacent PCBs **145** of the ECU **144** located within the housing **35**, extra wiring/cabling for the sensors **148.sub.a** can be reduced, if not eliminated (i.e. reduction and/or elimination of wires passing sensor signals **144.sub.f** through the housing **35** walls). For example, electromagnetic (EM) noise or interference is less likely to be picked up between the sensors components **148.sub.a** and the electronic components **148** since the mounting of the sensors **148.sub.a** via wires/electronic connections between the sensors **148.sub.a** and the PCB circuits is significantly shortened and therefore shielded by the housing **35** itself. As such, the location of the sensor components **148.sub.a** adjacent to the spindle (e.g. lead screw **140** with coupled extension member **40**), along with sensor signal processing capabilities via processing components **148** of the ECU **144**, provides for significant advantages as noted above.

(81) The foregoing description of the embodiments has been provided for purposes of illustration and description. It is not intended to be exhaustive or to limit the disclosure. Individual elements or features of a particular embodiment are generally not limited to that particular embodiment, but, where applicable, are interchangeable and can be used in a selected embodiment, even if not specifically shown or described. The same may also be varied in many ways. Such variations are not to be regarded as a departure from the disclosure, and all such modifications are intended to be included within the scope of the disclosure. Those skilled in the art will recognize that concepts disclosed in association with the example detection system can likewise be implemented into many other systems to control one or more operations and/or functions.

(82) Example embodiments are provided so that this disclosure will be thorough, and will fully convey the scope to those who are skilled in the art. Numerous specific details are set forth such as examples of specific components, devices, and methods, to provide a thorough understanding of embodiments of the present disclosure. It will be apparent to those skilled in the art that specific details need not be employed, that example embodiments may be embodied in many different forms and that neither should be construed to limit the scope of the disclosure. In some example embodiments, well-known processes, well-known device structures, and well-known technologies are not described in detail.

## Claims

1. A system for moving a closure panel between an open position and a closed position relative to a motor vehicle body, the system comprising: a latch for engaging a striker to maintain the closure panel in the closed position relative to the motor vehicle body; and at least one electromechanical biasing member comprising: an electric motor; and a slave controller; wherein a master controller is housed within the latch.
2. The system of claim 1, wherein: the master controller is a first controller; the slave controller is a second controller; the first controller is in communication with a vehicle control system; and the

- second controller is in communication with the first controller; wherein the first controller comprises at least one driver for generating an actuation signal for supply to a device to be controlled; wherein the second controller is configured to supply a command signal to the first controller.
3. The system of claim 2, wherein the second controller does not include a driver for generating the actuation signal for supply to the device to be controlled.
  4. The system of claim 3, wherein the driver of the first controller comprises a Field Effect Transistor.
  5. The system of claim 2, wherein the command signal is a high level command signal.
  6. The system of claim 2, wherein the second controller is configured in communication with at least one sensor.
  7. The system of claim 6, wherein the second controller is configured to communicate to the first controller sensor signals from the at least one sensor.
  8. The system of claim 2, wherein the second controller comprises a communication driver for communicating with the first controller using digital messages.
  9. The system of claim 8, wherein the digital messages are communicated over a vehicle network using a network protocol.
  10. The system of claim 9, wherein the network protocol includes one of a Local Interconnect Network Protocol and a Controller Area Network protocol.
  11. The system of claim 2, wherein the vehicle control system is a Body Control Module (BCM).
  12. The system of claim 2, wherein the first controller is fixed to a body of the vehicle.
  13. The system of claim 2, wherein the first controller is fixed to the closure panel.
  14. The system of claim 2, wherein the first controller is configured to receive a sensor signal and issue driver commands to the device on a door of the vehicle.
  15. The system of claim 14, wherein the sensor signal is an obstacle detection sensor signal.
  16. The system of claim 14, wherein the sensor signal is a switch signal.
  17. The system of claim 14, wherein the sensor signal is an accelerometer signal.
  18. A control system for a motor vehicle, comprising: a first controller in communication with a vehicle control system; and a second controller in communication with the first controller; wherein the first controller comprises at least one driver for generating an actuation signal for supply to a device to be controlled; wherein the second controller is configured to supply a command signal to the first controller; and wherein the first controller is mounted in a latch housing mounted to a door of the vehicle.
  19. A control system for a motor vehicle, comprising: a first controller in communication with a vehicle control system; and a second controller in communication with the first controller; wherein the first controller comprises at least one driver for generating an actuation signal for supply to a device to be controlled; wherein the second controller is configured to supply a command signal to the first controller; and wherein the first controller is mounted in a power actuator housing mounted to a door of the vehicle.
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